Dual Tetrad Gravity And The Field Origin
Deriving Fundamental Equations With $(3+1)^2$ Topology

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Abstract

The companion paper derived the gravitational constant G from spacetime topology with 0.12% accuracy and zero free parameters. This paper asks what else the same geometric structure determines.

The $(3+1)^2$ dual tetrad topology, two tetrahedra with opposite orientations coupled at a shared Field Origin, produces four quantitative results. (1) G from the coupling hierarchy $\alpha^{1+(3+1)+(3+1)^2}$, where changing the exponent by ± 1 shifts the prediction by a factor of 137. (2) The force hierarchy: gravity couples through 21 levels, electromagnetism through 1, giving $\alpha^{20} \approx 10^{-43}$. (3) The dark matter correction factor $\mu(x) = x/(1+x)$, derived from detailed balance in field closure, with characteristic acceleration $a_0 = k^2 G m_e / r_e^2$ (where $k = 4$ is the tetrahedral coupling number, m_e the electron mass, and r_e the classical electron radius) matching observation to 0.41%. This modifies gravity from $1/r^2$ to $1/r$ at galactic scales, explaining flat rotation curves without invisible mass. (4) The electron mass, from three overdetermined equations spanning gravity, cosmology, and electromagnetism, yielding m_e to 0.1% with no free parameters.

The topology also provides geometric interpretations of known physics: the inverse square law from three-dimensional field spreading, $E = mc^2$ from two closure speeds in the bimetric structure, spin $\hbar/2$ from the stellated octahedron's face ratio (2 of 8 faces point toward the coupling center), and energy quantization from Field Origin boundary conditions. The fine-structure constant is derived from the total interaction states of the stellated octahedron: $1/\alpha = 1 + 2k(k^2+1) + 1/[k(k+d)] = 137.036$, matching the measured value to 0.00021%.

The only external input is the Hubble constant H, a cosmological observable. The cosmological constant, strong force, and weak force are not addressed. The framework connects to established bimetric gravity through the Hassan-Rosen action in vielbein form, with the topology constraining the interaction parameters. Physical law is not arbitrary; behind the equations lies structure.

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I. INTRODUCTION

The fundamental constants of physics have been measured with extraordinary precision. Newton's gravitational constant G has been the subject of increasingly refined experiments since Cavendish's 1798 torsion balance measurement, yet for over three centuries no theory has calculated its value from first principles. The Standard Model of particle physics requires over two dozen such parameters [23], each determined by experiment but not derived from any deeper principle. That these constants take the values they do is what makes atoms stable, chemistry possible, and structure viable at every scale. Explaining why they take precisely these values has remained one of the deepest open questions in physics.

The stakes extend beyond unexplained constants. Galactic rotation curves, gravitational lensing, and the cosmic microwave background all indicate that approximately 85% of the universe's gravitating mass has never been directly detected [23]. Four decades of dedicated searches have produced no confirmed dark matter particle. The assumption underlying standard cosmology is that spacetime is a structureless continuum whose gravitational response is fully characterized by a single constant, G. But if the gravitational field possesses internal coupling structure that modifies its response at different scales, the question changes: is the missing 85% missing mass, or is it missing structure?

The consequences of these constants are visible at every scale, from atomic spectra to galactic rotation curves. In other branches of physics, the relationship between internal topology and observable properties is well established. The molecular lattice of a ruby crystal determines the discrete energy levels available to electrons passing through it; the result is stimulated emission, the physical basis of the laser. Semiconductor band gaps, superconducting transition temperatures, and phonon spectra all follow from the coupling architecture of their respective lattices. In each case, the internal topology of the medium determines the measurable constants that describe it. Gravity, too, possesses an internal coupling structure. The tetrad formalism [13, 14] describes it, and ghost-free bimetric gravity [10] extends it to two interacting metrics. Both are standard textbook material, taught in graduate courses worldwide. Yet the question of whether this coupling topology determines the gravitational constant, in the same way that lattice topology determines a crystal's properties, has not been systematically explored.

This paper pursues that question. Gravity occupies a unique position among the fundamental forces: it is not a field on a background spacetime but the dynamical spacetime itself, and it must therefore be self-consistent at every point with no external structure to enforce coherence. General relativity encodes this through the non-degeneracy of the metric ($\det(g) \neq 0$), which requires the field to *close*: to define a complete coupling structure at every point, with no gaps and no missing directions. Closure forces the field to develop internal architecture, creating the coupling chambers through which neighboring field regions interact. Each spacetime point carries a local frame of four basis vectors that couple to neighboring frames through shared boundary faces. The pattern of these couplings constitutes the field's topology. If this topology is constrained tightly enough by self-consistency, the fundamental constants may not be free parameters at all but consequences of how the gravitational field couples to itself.

Atomic structure emerges from electromagnetic field coupling: the interaction between charged particles produces persistent configurations with definite internal topology, and that topology determines measurable properties from spectral lines to chemical bonds. Gravitational field coupling produces structure by the same principle. When field regions interact through the (3+1) tetrad frames that general relativity already requires, the coupling creates a self-consistent field architecture whose topology is not a free choice but a consequence of the field's own consistency requirements.

The companion paper, "Deriving the Gravitational Constant from Spacetime Topology" [1], demonstrated that this topology is uniquely determined. The physical picture begins with field packing. Gravitational field regions, being distributed and locally coupled, pack in three dimensions like spheres. In close packing, each sphere touches 12 geometric neighbors, the kissing number for three dimensions [17]. But not all 12 contacts serve as active coupling channels. The simplex theorem establishes that $k = 4$ faces are both necessary and sufficient to enclose a volume in three dimensions, and these four faces define a tetrahedral chamber, the minimal solid enclosing the coupling region.

A single tetrahedral chamber, however, is not self-consistent. Four independent results from the literature, spanning five decades and over 2,300 combined citations, all demonstrate this. Spinor theory demands the SL(2,$\mathbb{C}$) double cover [7, 8]. Ashtekar's chiral decomposition requires both self-dual and anti-self-dual connections [9]. Ghost-free bimetric gravity requires

two dynamical metrics [10]. Stability analysis confirms that a single orientation produces a scalar ghost [6]. Self-consistency requires a second tetrahedron with opposite orientation, coupled to the first at a shared point: the interaction vertex [10, 11], which this paper identifies as the Field Origin. The resulting structure, two interlocking tetrahedra sharing a common center, is the stellated octahedron [5]. Just as an atom's internal topology determines its spectral and chemical properties, the stellated octahedron's coupling topology determines the integers in G.

The (3+1) that appears in this structure is not new. It is the same (3+1) that the ADM formalism uses to decompose four-dimensional spacetime into three spatial dimensions evolving through one temporal dimension, and the same (3+1) that tetrad gravity encodes in the four legs of the vierbein. The coupling topology of the gravitational field is built from the dimensional structure that these formalisms have described for decades. Applied to the stellated octahedron, it yields a quantitative prediction for G:

$$G = \frac{(3+1)^2+1}{3(3+1)+1} \times \alpha^{1+(3+1)+(3+1)^2} \times \frac{\hbar c}{m_e^2} \tag{1}$$

Topology: $G_{\text{topo}} = 6.666 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$

Measured: $G_{\text{CODATA}} = 6.674 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$ [22]

Agreement: 0.12%, zero adjustable parameters.

Every integer in Eq. (1) traces to one source: the dimensionality of space, which determines how the gravitational field couples to itself. The derivation follows a chain established in the companion paper [1]. Space has $d = 3$ dimensions. The simplex theorem requires $d+1 = 4$ boundary faces to enclose a volume in d dimensions, the same count that appears independently as the four constraint equations (one Hamiltonian, three momentum) in GR's ADM formalism [15]. This gives $k = (3+1) = 4$ coupling faces per field chamber, the (3+1) of spacetime. The field's coupling hierarchy then sums three levels:

$$1 + (3+1) + (3+1)^2 = 1 + 4 + 16 = 21 \tag{2}$$

The gravitational field traverses all 21 coupling levels, each attenuating by the fine-structure constant α, giving the exponent α^{21} in Eq. (1). The prefactor and the natural scale $\hbar c/m_e^2$ are developed in Section III after the topological structure has been fully established.

The chain $d = 3 \to (3{+}1) \to 1 + (3{+}1) + (3{+}1)^2 = 21 \to G$ has an internal structure that rewards closer examination. The hierarchy $1 + k + k^2$ is a polynomial in $k = (3{+}1)$:

$$f(k) = 1 + k + k^2 \tag{3}$$

This is not arbitrary notation. Each term represents a distinct level of the field's coupling hierarchy, and each connects to structures that are already recognized in gravitational physics:

- **The first term, 1:** The Field Origin, the coupling vertex at the center of the dual tetrahedron field chamber, where the two spacetime sectors share a common point. This is where $r = 0$ and both field sectors meet. It is the "1" in (3+1): the temporal dimension that makes three-dimensional space into spacetime. The Field Origin is always present in every field configuration; it is a topological fact about the structure, not a parameter.

- **The second term, $k = (3{+}1) = 4$:** One coupled face. When TetB couples to TetA, each face subdivides into $3 + 1 = 4$ sub-regions (three spatial, one coupling) through which the gravitational field propagates across distance (Figure 4). In the tetrad formalism, these four sub-regions correspond to the four legs of the vierbein [13]. This term represents the linear transport of the field through one sector of the dual tetrahedron field chamber. Crucially, field propagation requires spatial extent—distance ($r > 0$) and direction—making this the only term that responds to the structural state (Section III F).

- **The third term, $k^2 = (3{+}1)^2 = 16$:** The full bimetric field interaction. Both tetrahedra of the field chamber coupled at the Field Origin: the $4 \times 4 = 16$ components of the tetrad product $e^a{}_\mu e^b{}_\nu$, the same count that bimetric gravity recognizes as the total interaction channels between two metrics [10, 11]. This term represents the structural relationship between the two field sectors at their shared vertex. Unlike field propagation, this coupling is instantaneous and requires no distance.

Two of these terms, the Field Origin (1) and the bimetric field interaction (k^2), are properties of a point: they exist regardless of the field's structural state. The middle term (k), however, represents field propagation through spatial extent, which the coupling boundary at $r = 0$

does not possess. This structural asymmetry—the coupling hierarchy is recursive, with structure on either side of a non-structural boundary—will prove consequential when the topology of the Field Origin is examined in Section III C.

If this field topology derives G, does it also explain other fundamental relationships? The answer, developed in Sections IV through VII, is that the same dual tetrahedron field topology produces quantitative results across classical, relativistic, and quantum physics. The inverse square law follows from field spreading in three dimensions, the same dimensionality that determines the tetrahedral simplex. Mass-energy equivalence follows from the bimetric field structure and its two closure speeds. The force hierarchy, why gravity is 10^{43} times weaker than electromagnetism, follows from the depth of field coupling within the 21-level structure. Electron spin $\hbar/2$ follows from the face ratio of the stellated octahedron. The dark matter problem, why galaxies rotate faster than Newtonian gravity predicts, is resolved by the constitutive law that field closure imposes. These are not separate explanations for separate phenomena. They are aspects of one field topology, the topology that yields G.

A. Scope and Limits

This paper presents what the $(3 + 1)^2$ dual tetrad topology explains about fundamental physics. It does not claim to explain everything. The fine-structure constant α is derived in Section VI from the ratio of two measurable rates and from the total interaction states of the stellated octahedron, yielding $1/\alpha = 137.036$ to 0.00021%. The electron mass m_e is derived in Section VII F from an overdetermined closure system. The only external input is the Hubble constant H, a cosmological observable measured independently of particle physics. The cosmological constant, the strong force, and the weak force are not directly addressed. The goal is clarity about what the topology does explain, not a claim to have solved all of physics.

B. Overview

Section II establishes the theoretical foundations: LIGO confirms gravitational propagation, the wave equation requires neighbors, and field closure is what GR already enforces

through non-degeneracy. Section III reviews the $(3+1)^2$ geometric structure from which the derivations follow. Section IV derives the fundamental equations: G from tetrad coupling, the inverse square law, mass-energy equivalence, and the force hierarchy. Section V examines quantum consistency: energy quantization, wave-particle duality, and the energy-momentum relation. Section VI derives the fine-structure constant. Section VII resolves the dark matter problem, presents a precision table, and identifies the frontier ahead.

Prerequisites. This paper builds on results from two companion works. "Field Closure and the Topology of Gravity" [2] derives the $k = 4$ structure from the simplex theorem and establishes the three-route convergence on the electron radius. "Deriving the Gravitational Constant from Spacetime Topology" [1] develops the four independent arguments for dual tetrahedra and derives G to 0.12%. Key results are summarized but not re-derived here.

II. THEORETICAL FOUNDATIONS

This section establishes the theoretical basis for the derivations that follow, drawing on established results from gravitational physics. We distinguish between what is known in the literature and what this paper contributes. Readers familiar with the companion paper, "Deriving the Gravitational Constant from Spacetime Topology" [1], may proceed to Section III.

A. Gravity Propagates: From Metric to Topology

General Relativity describes gravity through the metric tensor $g_{\mu\nu}$, a dynamical field satisfying Einstein's equations. This is standard physics, confirmed by a century of observations. GR works beautifully. But when we ask how the gravitational field couples to itself, how one region of spacetime influences its neighbors, the answer lies in a reformulation that many physicists learn but few fully explore: tetrad gravity.

Tetrad gravity, also called vierbein or frame-field gravity, is textbook material [13, 14]. It describes the same physics as metric GR but makes the field's internal structure explicit. Each spacetime point has a local frame, a tetrad of four vectors, and these frames couple to neighboring frames. This coupling structure is the field's *topology*. The mathematics

has been in graduate textbooks for decades, yet the physical implications of this topology remain underappreciated.

Why does gravity, uniquely among the fundamental interactions, require us to ask about its internal structure? Because gravity is not a field that lives *on* spacetime. Gravity *is* spacetime. The electromagnetic field inherits its topology from the background metric: it knows what "neighbors" means because spacetime tells it. But the gravitational field is the metric itself. It cannot borrow structure from a background, because it is the background. If the field must be self-consistent and no external structure exists to enforce that consistency, the field must provide its own closure mechanism. This is the physical content of background independence, one of GR's deepest features.

This distinction has concrete consequences. Electromagnetic field lines are *open*: they begin on positive charges and end on negative charges. A photon propagates through the EM field without requiring the field to form a closed, self-consistent structure at every point. Gravity has no such freedom. General relativity demands that the metric be non-degenerate everywhere ($\det(g) \neq 0$), which means the gravitational field must *close*: it must define a complete, self-consistent geometry at every point, with no missing directions and no collapsed dimensions. A field that must close must hold volume, and holding volume requires structure, just as a house requires walls.

LIGO's first detection of gravitational waves in September 2015, which earned the 2017 Nobel Prize in Physics, confirmed that gravity propagates as a wave through locally-coupled structure. The subsequent detection on August 17, 2017, when LIGO and Virgo observed gravitational waves from a neutron star merger arriving 1.7 seconds before the gamma-ray burst [12], confirmed this structure with multi-messenger precision. The gravitational field has spatially distributed, locally-coupled degrees of freedom: the disturbance propagated from point to point, each region of the field influencing its neighbors across 130 million light-years.

The ocean makes this concrete. Water waves propagate because water molecules push their neighbors; the disturbance passes from molecule to molecule across thousands of kilometers. Water molecules must form closed electron shells, and this closure creates tetrahedral hydrogen-bonding structures, the same four-fold geometry that appears in tetrad gravity. Both water and gravity share a common pattern: the requirement that the field close forces

it to develop internal structure, and that structure determines how disturbances propagate.

These detections confirmed what tetrad gravity has described all along: local frames coupled to neighboring frames, field interacting with field, topology. The Nobel Prize recognized the detection of gravitational waves; the physical implication is that gravity has the distributed coupling structure that the closure requirement demands.

The wave equation encodes this requirement mathematically. Gravitational perturbations satisfy:

$$\frac{\partial^2 h}{\partial t^2} = c^2 \nabla^2 h \tag{4}$$

The Laplacian ∇^2 appears on the right-hand side. By definition, this operator requires evaluating the field at neighboring points:

$$\nabla^2 \Phi = \lim_{\epsilon \to 0} \frac{\Phi(x+\epsilon) + \Phi(x-\epsilon) - 2\Phi(x)}{\epsilon^2} \tag{5}$$

You cannot compute ∇^2 without knowing the field value at adjacent locations. This is not a physical assumption; it is the definition of the operator. The logic is then inescapable: gravitational waves propagate (LIGO observed this). Propagation requires ∇^2 (the wave equation demands it). ∇^2 requires neighbors (by definition). Therefore, gravity must consist of distributed, locally-coupled degrees of freedom. Figure 1 illustrates this schematically: each field region overlaps with its immediate neighbors, coupling through shared field volume, and the pattern of these couplings constitutes the field's topology.

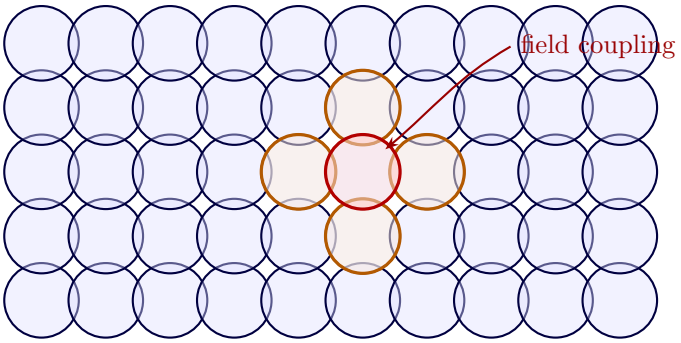


FIG. 1. Two-dimensional cross-section illustrating the coupling topology of gravitational field regions. Each circle represents a field region; where adjacent regions overlap (red shading), the fields couple through shared volume. One interior region (red) and its $k = 4$ nearest neighbors (orange) are highlighted, with the four coupling zones visible as lens-shaped intersections. In three spatial dimensions, the simplex theorem (Section II C) requires $k = d + 1 = 4$ coupling faces per region; these four coupled faces enclose a tetrahedral volume, the field chamber. The cross-section is shown in two dimensions for clarity; the essential feature, that field coupling creates enclosed topological structure, extends to the three-dimensional case.

This is not new physics. This is tetrad gravity, formalized decades ago and taught in graduate courses worldwide. What LIGO provided was empirical confirmation: the topology that tetrad theory describes is not merely mathematical convenience but physical reality.

The question is not *whether* gravity has topology; LIGO and tetrad theory both confirm it does. The question is: *what is* that topology?

B. Field Closure: The Requirement GR Already Enforces

If gravity consists of distributed, locally-coupled degrees of freedom, how do these degrees of freedom achieve consistency?

Consider what "locally-coupled" means. At each point, the field must define a local reference frame: which direction is timelike, which directions are spacelike. Neighboring points must agree on how their frames relate. If point A says "my timelike direction points this way" and neighboring point B says "that same direction is spacelike," physics would be undefined

at the boundary.

For the field to be well-defined everywhere, the local frames must form a complete, non-singular structure. At every point, all four directions (one time, three space) must be specified. No direction can be missing; no direction can be collapsed. This requirement, that the frame structure be complete everywhere, is what we call **field closure**.

This is not a new requirement. General Relativity has enforced it for a century, using different language. In metric GR, the condition is called non-degeneracy: $\det(g_{\mu\nu}) \neq 0$. Without this condition, the inverse metric $g^{\mu\nu}$ is undefined, indices cannot be raised, and the field equations cannot be written. Practitioners enforce non-degeneracy because without it, physics breaks. Field closure and non-degeneracy are two names for the same physics.

The tetrad formalism reveals what this condition means geometrically [13, 14]. The equivalence principle requires physics to be locally Minkowski at every point. This has a precise mathematical form:

$$g_{\mu\nu} = \eta_{ab}\, e^a_\mu\, e^b_\nu \tag{6}$$

where $\eta_{ab} = \mathrm{diag}(-1, +1, +1, +1)$ is the Minkowski metric and e^a_μ are four orthonormal basis vectors defining the local frame. The tetrad e^a_μ consists of one timelike vector e^0_μ and three spacelike vectors e^i_μ. These four vectors span the tangent space completely:

$$e^a_\mu\, e^\nu_a = \delta^\nu_\mu \tag{7}$$

Taking determinants of the tetrad equation:

$$\det(g) = -[\det(e)]^2 \tag{8}$$

Therefore $\det(g) \neq 0$ if and only if $\det(e) \neq 0$: the four tetrad legs must span the tangent space completely. **GR's non-degeneracy condition and tetrad closure are mathematically equivalent.**

Concept	**GR Language**	**Tetrad Language**
Field defined everywhere	Non-degenerate metric	Closure
Mathematical condition	$\det(g) \neq 0$	$\det(e) \neq 0$
Structure	3+1 decomposition	4 tetrad legs
Constraints	4 ADM equations	4 projections onto legs

TABLE I. GR non-degeneracy and tetrad closure are two names for the same requirement. Practitioners of General Relativity have enforced field closure for a century.

What does $\det(g) \neq 0$ mean physically? It means the metric must define distances in *all* directions. If $\det(g) = 0$, some direction is collapsed; the metric cannot describe motion in that direction; spacetime becomes singular. GR practitioners know this well: at event horizons, certain coordinate choices make $\det(g) \to 0$, and physics becomes ill-defined in those coordinates.

The tetrad formalism reveals the geometric content of this requirement. Non-degeneracy means the four basis vectors span all directions. This is closure: the field structure is complete, with no missing directions. The requirement practitioners have enforced for a century is precisely what tetrad gravity calls field closure.

C. Why $k = 4$: The Uniqueness of Four

Before examining the mathematics, consider what $k = 4$ means physically. The parameter k counts how many neighbors each field region must touch to achieve closure. In three-dimensional space, what is the minimum number of neighbors required to enclose a region?

Nature answers this question everywhere we look. Stack oranges at a grocery store: in close-packing each orange touches 12 neighbors, but the gaps between groups of three touching spheres form tetrahedral pockets, the simplest three-dimensional cavities. Examine water molecules: each forms hydrogen bonds with up to four neighbors, creating the tetrahedral network that gives water its remarkable properties. Study crystal structures: atoms pack in arrangements where tetrahedral coordination dominates. This is not coincidence. In 1611, Kepler conjectured that spheres in three dimensions pack most efficiently in arrangements

with tetrahedral local structure; Thomas Hales proved this rigorously in 1998 [17]. Four-fold coordination is not a choice. It is what three-dimensional space requires for efficient packing.

The simplex theorem [16] formalizes this observation: the minimum convex polytope enclosing volume in d dimensions has $d + 1$ faces. For $d = 3$ spatial dimensions, this gives 4 faces: a tetrahedron. The tetrahedron is the simplest possible three-dimensional container. Just as a triangle is the minimum polygon in 2D (you cannot enclose area with fewer than 3 sides), a tetrahedron is the minimum polyhedron in 3D (you cannot enclose volume with fewer than 4 faces).

When gravitational field regions couple to achieve closure, they must touch enough neighbors to form a complete structure. In three dimensions, that minimum is four. This is why $k = 4$.

Now the technical content follows naturally. The tetrad has four legs: one timelike, three spacelike. The ADM formalism [15] yields four constraints: one Hamiltonian, three momentum. LIGO measured exactly two graviton polarizations [12]. The symmetric metric tensor in four dimensions has 10 independent components; subtracting $2k$ constraints leaves $10-2k$ propagating degrees of freedom. For $k = 4$, this gives $10 - 8 = 2$, matching observation. The number four appears everywhere because three-dimensional space demands it.

The value $k = 4$ is uniquely determined by two bounds:

Why not $k < 4$? Fewer than 4 constraints cannot close a 3D evolution problem. The simplex theorem establishes this geometrically: minimal enclosure in d dimensions requires $d + 1$ boundary conditions. For $d = 3$ spatial dimensions, $k \geq 4$. Attempts to formulate gravity with fewer constraints fail to propagate consistently.

Why not $k > 4$? More than 4 constraints overconstrain the system. With $k > 4$, the equation $10 - 2k < 2$ yields fewer than 2 propagating degrees of freedom. Theories with $k = 5$ or higher either have no propagating modes (violating the observation that gravity propagates) or produce Boulware-Deser ghost instabilities [18].

Combining the bounds: $k \geq 4$ (closure) and $k \leq 4$ (stability). The only solution is $k = 4$. This is confirmed observationally by LIGO's measurement of exactly 2 graviton polarizations, consistent with $10 - 2(4) = 2$.

The $k = 4$ structure is not a hypothesis. It is the dimensional requirement of 3+1 spacetime: four directions (one time, three space), four tetrad legs, four ADM constraints, four faces

on the minimal simplex. From oranges to water molecules to gravitational fields, four-fold coordination is what three-dimensional space produces.

D. The Characteristic Scale: Where the Tetrad Lives

The tetrad formalism specifies topology ($k = 4$ faces) but not scale. At what physical radius does this internal structure exist?

Two independent routes converge on the same characteristic length:

Route 1: Electromagnetic Stability. An electron carries charge, and that charge creates an electric field extending outward in all directions. This field contains energy: the closer you pack electric field lines together, the more energy is stored. Now imagine shrinking the electron to a smaller and smaller radius. The field becomes more concentrated, and the stored electromagnetic energy increases.

At some critical radius, the energy stored in the electron's electric field equals its rest mass energy $m_e c^2$. This is a natural boundary: shrink the electron further, and the field energy would exceed the mass energy, violating energy conservation. The radius where these two energies balance is the classical electron radius [19]:

$$r_e = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} = 2.818 \times 10^{-15} \text{ m} \tag{9}$$

This calculation uses only electromagnetism (e, ε_0) and special relativity (c). No gravity. No cosmology. Yet it defines a characteristic scale for the electron based purely on electromagnetic self-consistency.

Route 2: Gravitational Topology. When gravity couples through the $(3+1)^2$ structure, all $k^2 = 16$ coupling channels engage (this factor will be derived in Section III E). The topologically-coupled gravitational acceleration at radius r is therefore $a = k^2 Gm/r^2$.

But at what radius does this internal structure stabilize? Nature provides a natural scale: the acceleration $a_0 \approx 1.2 \times 10^{-10}$ m/s^2, first identified empirically by Milgrom [21] from galaxy rotation curves. This acceleration marks the threshold where gravitational dynamics change character, appearing throughout astrophysics from dwarf galaxies to galaxy clusters. Remarkably, $a_0 \approx cH/(2\pi)$, linking it to the Hubble expansion rate H, the scale of the observable universe.

The characteristic radius emerges when we ask: at what distance from the electron does the tetrad's gravitational acceleration equal this cosmic threshold? Setting $a = k^2 G m_e / r^2 = a_0$ and solving for r:

$$r_e = k\sqrt{\frac{G m_e}{a_0}} = 2.82 \times 10^{-15} \text{ m} \tag{10}$$

This is the radius where the tetrad's internal gravitational coupling meets the cosmic acceleration scale. The calculation uses gravity (G), cosmology (a_0), and topology ($k = 4$). No electromagnetism.

Route	**Constants used**	**Result**	**Domain**
EM stability	e, ε_0, m_e, c	$r_e = 2.818$ fm	Electromagnetism
Cosmic topology	G, H, m_e, k, c	$r_e = 2.82$ fm	Gravity + Cosmology

TABLE II. Two independent routes converge on the same scale to 0.21%.

Match: **0.21%**. Two independent domains of physics, using different constants and different reasoning, converge on the same scale. This convergence is unique to the electron among known particles.

The convergence is not coincidence; it is constraint. There is exactly one scale where both routes agree. This is the characteristic size of the gravitational field's internal structure, the "horizon" within which the tetrad topology exists. The electron radius is not merely an electromagnetic constant; it is the tile size at which gravity's topological structure becomes manifest.

E. The Dual Tetrahedron Field Chamber

Let us take stock of what the previous sections established. LIGO confirmed that gravity propagates as waves through locally-coupled field regions, not through a structureless continuum. The simplex theorem established that these field regions pack with $k = 4$ coordination, and their coupling dynamics produce tetrahedral field chambers, the minimum structure capable of enclosing volume in three dimensions. Two independent routes, electromagnetic and gravitational, converged on the same characteristic scale for this field chamber: the electron radius $r_e \approx 2.82$ fm.

We now know three things: the gravitational field consists of distributed, locally-coupled regions; these regions pack tetrahedrally, and their field coupling produces a definite topology; and the characteristic size of each field chamber is the electron radius. What does this field topology look like?

From field coupling to the dual tetrahedron field chamber. Return to the picture of gravitational field regions packing like spheres. In optimal packing, each sphere touches 12 neighbors, the kissing number for three dimensions proven by the Kepler conjecture [17]. But not all 12 contacts are field coupling interfaces. Section II C established that field closure requires exactly $k = 4$ coupling channels. Of the 12 geometric neighbors, only 4 serve as active field coupling ports through which the field achieves closure; the remaining 8 contacts are geometrically present but do not participate in the field closure process. These 4 active channels define the faces of a tetrahedral field chamber, the minimal solid enclosing the field coupling region.

The tetrahedral coordination that field coupling produces is consistent with, and indeed demands, a dual structure. Each field chamber has an orientation, an apex pointing in some direction. But four independent arguments (below) establish that a single orientation is unstable. A second tetrahedron with opposite orientation must couple to the first, forming a pair. These two tetrahedra interpenetrate, sharing a common center where the fields couple, the point this paper identifies as the Field Origin. The resulting structure is the stellated octahedron [5]: two tetrahedra interlocking so that each vertex of one lies at the center of a face of the other.

Picture this: take a tetrahedron and place it apex-up. Now take a second tetrahedron, rotate it 180°, and lower it so the two share the same center. The eight vertices (four from each) now form the corners of a stellated octahedron. The two tetrahedra are distinct but field-coupled, opposite in orientation but coupled at the Field Origin.

Why dual structure is required. Four independent arguments from textbook physics, spanning 50 years and over 2,300 combined citations, establish that a *single* tetrahedral field chamber cannot describe gravity consistently:

1. **Stability** (Alexander, Marcianò, Smolin 2014 [6]): A single tetrahedral field chamber has a preferred direction, an apex pointing somewhere. This asymmetry produces what

physicists call a *ghost*: a field mode with negative kinetic energy that destabilizes the vacuum. Two opposite field chambers cancel the asymmetry, and the ghost disappears.

2. **Spinor structure** (Weinberg 1995 [7]; Geroch 1968 [8]): Fermions require the SL(2,C) double cover of the Lorentz group. In physical terms: rotating a fermion 360 degrees does not return it to its original state; a full 720-degree rotation is needed. This is bimetric geometry: one 360-degree rotation brings TetA back into alignment but leaves TetB out of phase; the second rotation realigns both. The double rotation IS the dual field structure. Each tetrahedron realizes one sheet of the double cover, one sector of the coupled field.

3. **Chirality** (Ashtekar 1986 [9]): Gravity decomposes into self-dual and anti-self-dual components, $A^+ \oplus A^-$, like left-handed and right-handed gloves that are mirror images but cannot be superimposed. Full gravitational field closure requires both halves. Each half corresponds to one tetrahedron, one sector of the field.

4. **Bimetric gravity** (Hassan, Rosen 2012 [10]): Ghost-free massive gravity requires two dynamical metrics. Each metric corresponds to one tetrahedral field chamber, and the two fields must couple at a shared point, the Field Origin.

These are textbook results from different subfields: stability analysis, quantum field theory, canonical gravity, and massive gravity. All four independently require the same field topology: two tetrahedra with opposite orientations, coupled at a shared center, the Field Origin. This is precisely the dual tetrahedron field chamber that $k = 4$ field coupling produces.

The ghost problem resolved. The stability argument deserves emphasis because it explains why the dual field structure is physically necessary. A single tetrahedral field chamber's asymmetry creates a ghost, a field mode with negative kinetic energy. Negative kinetic energy means the vacuum becomes unstable: small fluctuations grow without bound, creating particle-antiparticle pairs from nothing. This ghost problem plagued massive gravity research for over 70 years, from 1939 to 2010 [18].

The resolution lies in the field topology. Two tetrahedral field chambers with opposite orientations cancel the asymmetry: where one apex points up, the other points down. The preferred field directions cancel, and the ghost disappears. The stellated octahedron is not

merely a pretty shape; it is the unique field chamber configuration that eliminates the ghost while satisfying all four textbook requirements.

The dual tetrahedron field chamber is not new physics: it is the structure that field coupling produces when the gravitational field closes in three dimensions, and it is what GR requires for stability.

Summary. The theoretical foundations are established in the literature: the gravitational field propagates (LIGO), propagation requires field topology (wave equation), field topology requires field closure (non-degeneracy), field closure is $k = 4$ (uniqueness), the scale is the electron radius (convergence), and a single tetrahedron is unstable (four roads). Field coupling in three dimensions produces the dual tetrahedron field chamber. What remains is to show what this field topology produces. The following sections derive fundamental equations from this single topological foundation.

III. THE $(3+1)^2$ FOUNDATION

Section II established that gravity has topological structure: fields propagate, propagation requires neighbors, and consistency requires closure. The closure topology is $k = 4$, the characteristic scale is the electron radius, and stability requires two tetrahedra. This section develops the geometric details of that structure, establishing notation used throughout. Readers familiar with "Deriving the Gravitational Constant from Spacetime Topology" [1] may proceed to Section IV.

A. What the Dual Requirement Reveals

Section II E established that four independent arguments require dual tetrahedra. The question now is not *whether* two tetrahedra are required, but *what geometric constraints* these requirements impose.

Each argument tells us something specific about the structure:

Stability constrains *orientation*: the tetrahedra must point in opposite directions. Where one apex points up, the other must point down. This is not a choice but a requirement; any

other arrangement reintroduces the ghost.

Spinor structure constrains *topology*: the two tetrahedra cannot be the same structure traversed twice. They must be genuinely distinct, realizing the two sheets of the SL(2,C) double cover. This is why fermions can exist.

Chirality constrains *completeness*: each tetrahedron carries one half of gravity, the self-dual (A^+) or anti-self-dual (A^-) component. Both are required for the field equations to be complete.

Bimetric structure constrains *interaction*: the two tetrahedra must couple at a shared point. Two separate tetrahedra floating independently would not produce a consistent gravitational field. They must meet.

Together, these constraints uniquely determine the geometry: two tetrahedra, opposite orientations, sharing a common center. The next subsection identifies the only arrangement satisfying all four requirements.

B. The Stellated Octahedron

Given two tetrahedra with opposite orientations, how should they be arranged? The answer is uniquely determined: the stellated octahedron [5].

Picture two tetrahedra, one pointing up and one pointing down, interpenetrating so that each vertex of one lies at the center of a face of the other. The resulting figure has:

- 8 vertices (4 from each tetrahedron)
- 12 edges (6 from each tetrahedron)
- 8 triangular faces (4 from each tetrahedron)
- A central octahedral region where the two tetrahedra overlap
- Perfect central symmetry: rotating 180° about any axis through the center maps the structure onto itself

These 28 geometric elements ($8 + 12 + 8$) constitute the complete topological scaffold of the structure. This count will prove significant when the coupling hierarchy is read for its total breadth rather than its depth (Section VI).

This is the unique compound of two tetrahedra that satisfies all four requirements from Section III A. Other arrangements (tetrahedra side by side, apex to apex, or partially overlapping) either break the central symmetry required by bimetric gravity or fail to provide the complete spinor doubling. Only the stellated octahedron has both the dual orientation needed for stability and the shared coupling point needed for bimetric interaction. The structure is not chosen; it is required.

Figure 2 shows this geometry. The two tetrahedra interpenetrate completely: TetA (thick lines) points upward with its apex at the top, while TetB (thin lines) points downward with its apex at the bottom. At the geometric center, equidistant from all eight vertices, lies the point where the two structures couple. This coupling point, marked as the Field Origin, is where the bimetric interaction occurs. The next subsection develops the physics of this special point.

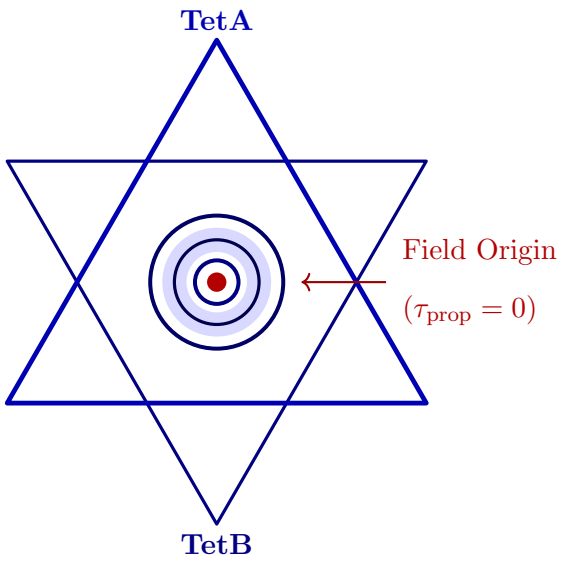


FIG. 2. The stellated octahedron: two tetrahedra with opposite orientations sharing a common center. TetA (thick lines) points upward; TetB (thinner lines) points downward. The structure has 28 geometric elements: 8 vertices, 12 edges, and 8 faces (4 of each per tetrahedron). At the geometric center lies the Field Origin, where the two spacetime sectors couple. This is the $(3+1)^2$ topology from which the fundamental equations derive.

C. The Field Origin

Bimetric gravity requires that two dynamical metrics interact. But how do they interact? The Hassan-Rosen formulation [10] and the vielbein extension by Deffayet, Mourad, and Zahariade [11] establish that the two spin-2 fields must couple at a shared spacetime point, what the literature calls an *interaction vertex*. Without such a vertex, the two metrics would be causally disconnected: two separate gravitational fields with no physical relationship.

The stellated octahedron provides exactly this interaction vertex. At the geometric center, equidistant from all eight vertices, the two tetrahedra share a single point. This is where the two spacetime sectors meet, where information from one tetrahedron can couple to the other. We call this point the **Field Origin**.

Why does time vanish at this point? Consider the light-travel time from the coupling center to any point at distance r. In the local tetrad frame, where spacetime is Minkowski, the propagation time along a radial null geodesic is:

$$\tau_{\text{prop}} = \frac{r}{c} \tag{11}$$

At the center itself, $r = 0$:

$$\tau_{\text{prop}} = \frac{0}{c} = 0 \tag{12}$$

The propagation time vanishes at the Field Origin. But this equation says more than "time is zero here." Read in the other direction, $\tau_{\text{prop}} = r/c$ means time *increases* with distance from the origin. Time radiates outward. The Field Origin is not merely a point where time happens to vanish; it is the source from which all time structure emerges. The analogy is the eye of a hurricane: perfectly still, yet generating the entire storm. The Field Origin has $\tau_{\text{prop}} = 0$ while being the point from which temporal separation radiates into both tetrahedral sectors.

This property is what makes the interaction vertex physically meaningful. The two tetrahedra represent two separate spacetime sectors with opposite orientations. For ghost-free coupling, they must meet at a point where their interaction can be instantaneous, unmediated by propagation delay. The Field Origin, where $\tau_{\text{prop}} = 0$, satisfies this requirement. It is the unique point where the two sectors can exchange information without temporal separation.

Why everything spins

One of the deepest questions in physics: why does everything rotate? Stars spin. Galaxies spin. Electrons spin. Angular momentum pervades the universe at every scale. Is this a coincidence, or does it follow from something fundamental?

Einstein-Cartan theory, developed in 1922, provides a hint [20]. Cartan showed that spacetime geometry and intrinsic spin are intimately connected: the spin of matter couples directly to spacetime torsion. In this framework, spin is not merely a property that particles happen to have; it is a geometric feature of spacetime itself. Where there is torsion, there is spin. Where there is spin, there is torsion.

The dual tetrahedral field structure makes this connection concrete. When two tetrahedra with opposite orientations couple at a shared point, they create a continuous rotational field topology. Angular momentum propagates through both sectors, and the opposite orientations ensure it flows through the coupling point rather than stopping there. The field structure does not merely permit rotation; it *requires* it.

The vortex topology

The Field Origin has a distinctive field topology: it is a vortex core. Consider the analogy of water flowing in a river whirlpool (Figure 3, left panel). At the surface, water spirals inward. As it approaches the center, it accelerates, passes through the central axis, and disperses below. The vortex core is not where motion stops; it is where rotation vanishes but through-flow continues.

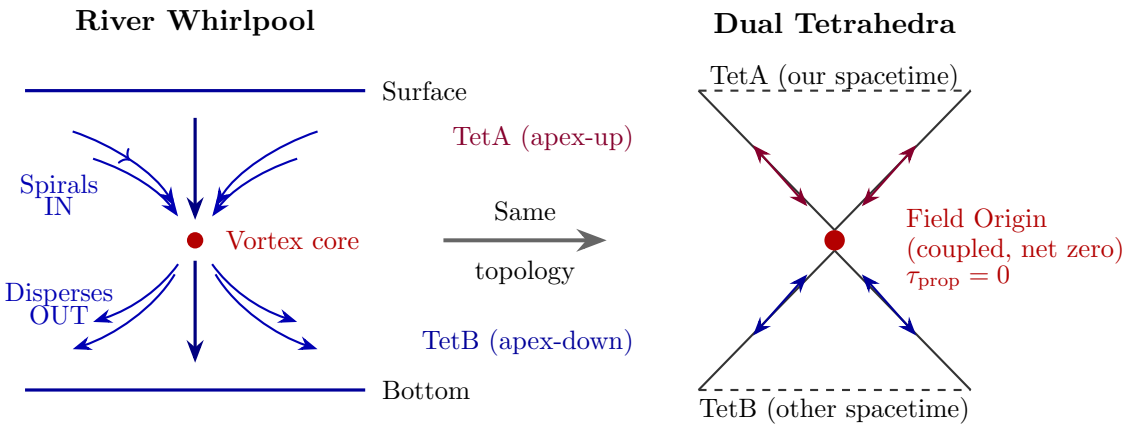


FIG. 3. The Field Origin as a vortex core. **Left:** A river whirlpool viewed from the side. Water spirals inward at the surface, passes through the vortex core, and disperses below. **Right:** The dual tetrahedral field structure. TetA spacetime (purple, top) with apex-up orientation, the Field Origin (red, center) at the coupling point where $\tau_{\text{prop}} = 0$, and TetB spacetime (blue, bottom) with apex-down orientation. Double-headed arrows indicate that the field propagates through each sector. The topology ensures continuity: angular momentum does not stop at the Field Origin but flows through it.

The dual tetrahedral field structure shares this topology (Figure 3, right panel). Angular momentum propagates through both spacetime sectors, passing through the Field Origin that connects them. TetA (apex-up) and TetB (apex-down) have opposite field orientations, but the topology ensures continuity: the field does not stop at the coupling point but flows through it. At the Field Origin itself, the rotational component vanishes (because $\tau_{\text{prop}} = 0$ there), yet angular momentum continues through. This is the geometric origin of spin: angular momentum flowing through the topological eye of the vortex.

How much angular momentum passes through this vortex eye? The answer depends on the internal face structure of the stellated octahedron, which determines the aperture of the vortex core.

D. The Face Structure

Section III C established that angular momentum flows through the Field Origin, the vortex eye of the dual tetrahedral field structure. How much passes through depends on the

aperture: the internal face structure of the stellated octahedron that filters what enters the vortex core.

Each tetrahedron has 4 triangular faces. The stellated octahedron, composed of two interlocking tetrahedra, therefore has 8 faces in total. These 8 faces divide into two types based on their orientation relative to the Field Origin:

- **TIME faces** (2 total): In each tetrahedron, the face opposite the apex points inward, toward the geometric center where the Field Origin lies. The reader can verify this on Figure 2: the base of TetA (apex-up) faces downward toward the center, and the base of TetB (apex-down) faces upward toward the center. These two faces point toward the coupling point where $\tau_{\text{prop}} = 0$. They are called TIME faces because they connect to the temporal origin.

- **SPATIAL faces** (6 total): The remaining three faces of each tetrahedron share the apex vertex and point outward, away from the Field Origin into the spatial dimensions. Three faces from TetA radiate upward and outward; three faces from TetB radiate downward and outward.

Why exactly 2 faces point inward is a geometric fact, not a parameter. A tetrahedron has one apex and one base. The base, the single face opposite the apex, points toward the center of the stellated octahedron. The other three faces share the apex and point away. With two tetrahedra, two bases point inward and six apex-sharing faces point outward. The ratio 2:6, or equivalently 2 of 8 total faces oriented toward the Field Origin, is fixed by the topology of the stellated octahedron.

This ratio will have physical consequences. The vortex topology of Section III C established that angular momentum flows through the Field Origin. The face structure establishes that only 2 of the 8 faces serve as the aperture for this flow. These faces determine the observable fraction of the total field coupling. To know the total, we need the coupling hierarchy: how many field coupling levels operate within the stellated octahedron. Once we have that hierarchy, the face ratio and the polynomial will combine to determine spin.

E. The Coupling Hierarchy

Section III D established the face structure of the stellated octahedron: 8 faces, of which 2 point inward toward the Field Origin and 6 point outward. These faces are the aperture through which the gravitational field couples. But how deep does the coupling go? The answer is a polynomial that counts the total number of field coupling levels within the stellated octahedron.

The structure $(3+1)$ denotes three spatial dimensions plus one time dimension: ordinary spacetime. With two coupled tetrahedra, we have $(3+1)^2$: two spacetimes interacting.

But field coupling between the two tetrahedra does not happen all at once. It has levels, each contributing to the total interaction strength. Consider what must connect for gravity to operate:

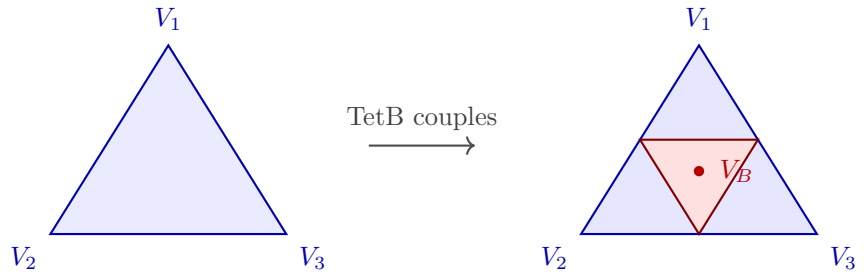
1. **The Field Origin itself (contributes "1").** Field coupling begins at the origin, where the two tetrahedra meet. This is the foundation: without the Field Origin, there is no coupling at all.

2. **One coupled face (contributes "4").** From the Field Origin, the field propagates through one face of the tetrahedron, which subdivides into $3+1=4$ sub-regions (three spatial, one coupling) as shown in Figure 4. This is how the field extends through one sector of the dual tetrahedron field chamber.

3. **The full bimetric interaction (contributes "16").** Both tetrahedra interact fully. Each has (3+1) dimensions, so the complete field coupling space is $(3+1)\times(3+1)=16$-dimensional.

The total field coupling hierarchy is a polynomial in $k=(3+1)=4$:

$$f(k) = 1 + k + k^2 = 1 + (3+1) + (3+1)^2 = 1 + 4 + 16 = 21 \tag{13}$$



(a) Face of TetA **(b)** Coupled face: $3 + 1$

FIG. 4. The coupling topology of a single face. **(a)** A triangular face of TetA. **(b)** When TetB couples through the Field Origin, its vertex V_B penetrates the face center, subdividing the face into $3 + 1 = 4$ sub-regions: three corner sub-regions (blue) anchored at the original vertices, and one central coupling sub-region (red) at the penetration point. This is the same $3 + 1$ structure that appears within each tetrahedron (3 spatial +1 temporal face) and across both tetrahedra ($(3+1)^2 = 16$ interpenetrations). The polynomial's hierarchy is the geometry of this single coupled face, recurring at every scale.

The gravitational field traverses all 21 coupling levels: the Field Origin (1), one coupled face ($k = 4$), and the full bimetric interaction ($k^2 = 16$). Each level attenuates by the fine-structure constant α, giving the coupling depth α^{21} in the expression for G. The number 21 is derived from the field topology, not imported from elsewhere; the polynomial $f(k) = 1+k+k^2$ counts the levels and the topology determines $k = 4$.

Read as a narrative, the hierarchy tells a story: begin at the Field Origin (the "1"), from which the field propagates outward through one coupled face ($(3 + 1)$), and through which the two sectors couple to each other ($(3 + 1)^2$). Gravity must traverse this entire chain. Without the Field Origin, there is no coupling. Without coupling, there is no face through which the field can propagate. Without two sectors, there is no bimetric interaction. The exponent 21 counts every channel in this chain.

Clarification: the "1" versus the "+1"

The hierarchy contains a "1" and the spacetime structure contains a "+1." These are related but distinct.

The "+1" in $(3 + 1)$ represents time as a *dimension*. It exists everywhere in spacetime; every point in the universe has a time coordinate. This is what distinguishes spacetime from ordinary three-dimensional space.

The standalone "1" in the hierarchy represents time's *origin*, the specific point where the time coordinate equals zero. The distinction is analogous to "the x-axis" (a dimension extending everywhere) versus "$x = 0$" (a specific point on that axis).

Symbol	**What it represents**	**Where it applies**
The "+1" in $(3 + 1)$	Time as a dimension	Throughout spacetime
The "1" in the hierarchy	Time's origin	At $r = 0$, where $t = r/c = 0$

TABLE III. The distinction between time as dimension and time's origin.

The companion paper [1] identified the "1" with time. This identification was correct; the present clarification specifies: the "1" is where time equals zero.

Why the hierarchy explains $E = mc^2$

The field coupling hierarchy not only counts channels; it explains why Einstein's equation takes the form $E = mc^2$.

Consider where mass exists in this field structure. Mass is energy that does not propagate: a photon (massless) travels at c, while a massive particle at rest remains localized. In the dual tetrahedral framework, mass is energy bound at the Field Origin, the "1" where the two tetrahedra couple and where $\tau_{\text{prop}} = 0$. The binding mechanism is topological: energy at the field coupling point is shared between the two tetrahedra simultaneously, with each sector's boundary conditions preventing the energy from propagating outward through either one alone. A particle with mass is a stable configuration of energy localized at this field coupling point.

Now consider what happens during mass-energy conversion. Energy stored at the Field Origin must propagate outward. But outward through what? Through both tetrahedra. The structure is $(3 + 1)^2$, not $(3 + 1)$. Two spacetimes, not one.

Each tetrahedron propagates disturbances at the closure speed, the speed at which field consistency is maintained. Lorentz invariance requires this speed to equal c: any finite propagation speed for a fundamental field must be the invariant speed, since a frame-dependent closure speed would violate the equivalence principle. The c speed limit is therefore not a traffic law imposed on objects moving through space; it is the processing speed of the spacetime structure itself. Exceeding c would mean outrunning the very field that defines one's existence. When energy leaves the Field Origin, it engages both:

$$E = m \times c_{\text{TetA}} \times c_{\text{TetB}} = m \times c \times c = mc^2 \tag{14}$$

The "squared" in c^2 has the same origin as the "squared" in $(3+1)^2$: both reflect the bimetric structure. This provides a geometric interpretation of why mass-energy conversion involves c^2 rather than some other power. The standard derivation from Lorentz invariance establishes that c^2 is required; the $(3+1)^2$ topology suggests *why* spacetime has the structure that produces this requirement.

The hierarchy thus answers three questions simultaneously:

1. **Where is mass?** At the "1" (the Field Origin, where $t = 0$)
2. **Why does conversion involve c?** Because each tetrahedron has closure speed c
3. **Why c^2?** Because $(3+1)^2$ means two tetrahedra, hence $c \times c$

Einstein's equation is not separate from the coupling hierarchy. It is a direct reading of the $(3+1)^2$ field structure: mass at the Field Origin, conversion through two spacetimes, c^2 from two closure speeds.

The polynomial $f(k) = 1 + k + k^2 = 21$ counts the total field coupling levels when all channels are active. But the gravitational constant contains three integers: 21, 17, and 13. Where do the other two come from? The answer lies in the three structural regions of the stellated octahedron. The coupling hierarchy follows a recursive pattern—structure, origin, structure—and the structural state coefficient $s = +1, 0, -1$ determines whether the face channels that carry the field contribute, vanish, or reverse.

F. The Three Structural States

The 21 coupling levels of Section III E assume all channels are active. In the actual stellated octahedron, however, the coupling hierarchy is recursive: structure couples through origin to structure. Different structural regions see different channels because the origin is the non-linear coupling boundary between structures, not a structural region itself.

Of the polynomial's three terms, each maps onto a level of the recursive structure. The Field Origin (1) is the non-linear coupling boundary: the point where $r = 0$ and $\tau_{\text{prop}} = 0$, present in every field configuration. The bimetric interaction ($k^2 = 16$) is a relationship between two structures at that boundary, the 4×4 interaction between two tetrads: if TetA couples to TetB, TetB couples to TetA, requiring no preferred direction. The middle term, $k = 4$, is different in kind: it represents field propagation through the four channels of one coupled face, the linear transport of the field through one sector of the stellated octahedron. Field propagation requires spatial extent—distance ($r > 0$) and direction—and therefore exists only within a structural region, never at the coupling boundary. The Field Origin excludes face channels not as an exception but as a structural necessity: the coupling boundary between structures cannot itself contain structure without breaking the recursive hierarchy.

The stellated octahedron has three distinct structural regions, and the state coefficient s determines whether face channels contribute:

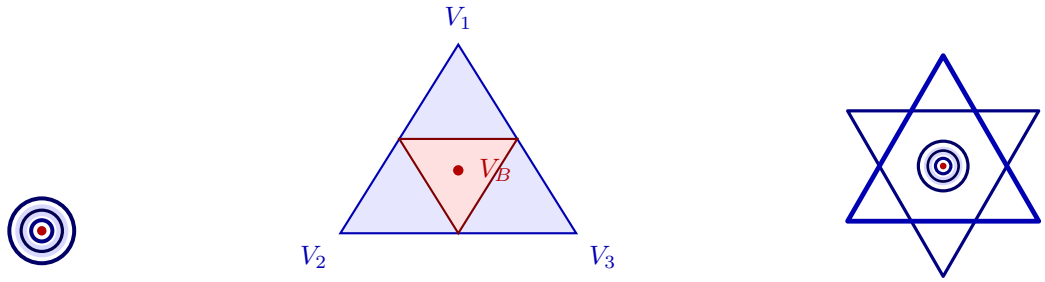
- In **TetA** ($s = +1$), the field propagates outward through the four face channels of its coupled face with full coupling. The middle term contributes $+k$.
- At the **Field Origin** ($s = 0$), the two tetrahedra couple at $r = 0$. This is the non-linear coupling boundary, not a structural region: no spatial extent exists for face channels to operate. The middle term contributes 0.
- In **TetB** ($s = -1$), the field propagates outward through the opposite sector's coupled face. The reversed orientation inverts the face contribution. The middle term contributes $-k$.

These are the three structural states of the dual tetrad: three regions of the stellated octahedron where the polynomial evaluates to three different values. The states determine *where*

the polynomial is evaluated; the three terms of the polynomial are the geometric *components* from which each evaluation is built.

Recall the chain $d = 3 \to (3{+}1) \to 1 + (3{+}1) + (3{+}1)^2 = 21 \to G$ has an internal structure. The hierarchy $1 + k + k^2$ is a polynomial in $k = (3{+}1)$, Eq. (13), whose three terms map onto the three geometric components of the stellated octahedron:

Term	**Value**	**Panel**	**Geometric component**
1	1	(a)	Field Origin — non-linear coupling boundary at $r = 0$
k	4	(b)	One coupled face — 4 channels of field propagation
k^2	16	(c)	Full stellated octahedron — 16 bimetric interaction channels



(a) **(b)** **(c)**

FIG. 5. The three geometric components of $f(k) = 1 + s \cdot k + k^2$: **(a)** the Field Origin (1), **(b)** one coupled face ($k = 4$), and **(c)** the full stellated octahedron ($k^2 = 16$). The structural state s determines which components contribute to the polynomial's evaluation.

Evaluating $f(k) = 1 + s \cdot k + k^2$ in each structural state gives the three integers in G:

Structural state	**Polynomial**	**Value**	**Role in G**
TetA ($s = +1$, apex-up)	$1 + (3{+}1) + (3{+}1)^2$	$= 21$	exponent
Field Origin ($s = 0$)	$1 + \quad 0 \quad + (3{+}1)^2$	$= 17$	numerator
TetB ($s = -1$, apex-down)	$1 - (3{+}1) + (3{+}1)^2$	$= 13$	denominator

All three integers in G emerge from one polynomial evaluated in the three structural regions of the stellated octahedron.

Why only the middle term changes

The structural argument is direct. The coupling hierarchy is recursive: structure couples through origin to structure. The face channels (k) are the only term that represents propagation through space; the origin (1) and the bimetric interaction (k^2) are properties of a point. Including structure within the coupling boundary would break the recursion—the boundary must remain a pure non-linear coupling point to mediate between the two structural sectors. The face channels are absent at $s = 0$ not as an exception, but as the condition that preserves the hierarchy.

Parity provides the mathematical confirmation. The polynomial $f(k) = k^0 + k^1 + k^2$ contains three terms with powers 0, 1, and 2. Under orientation reversal ($k \to -k$):

- $k^0 = 1 \to (-k)^0 = 1$ (even power, unchanged)
- $k^1 = k \to (-k)^1 = -k$ (odd power, sign reverses)
- $k^2 = 16 \to (-k)^2 = 16$ (even power, unchanged)

The two even-powered terms are immune to orientation reversal. Only the single odd-powered term, k^1, changes sign. Physically: a point (k^0) has no spatial extent to reverse. A relationship (k^2) is symmetric; if TetA couples to TetB, then TetB couples to TetA regardless of which is called "apex-up." But propagation (k^1) has direction and spatial extent, and reversing the field orientation reverses that direction. The ghost-free condition fixes two opposite orientations (Section III A), plus one boundary where they couple ($r = 0$), giving exactly three structural states. No fourth configuration exists. One polynomial, three regions, three integers.

The prefactor: coupling capacity over spatial structure

The three structural states determine where each integer appears in G. The exponent 21 counts the full coupling depth from TetA (apex-up), where all channels are active. But the prefactor is a ratio: it compares two structural states. The question is which two, and why.

The numerator, 17, is the polynomial evaluated at the Field Origin ($s = 0$). The algebraic structure reveals what this counts: $17 = k^2 + 1 = (3{+}1)^2 + 1$. At the coupling boundary,

face channels are absent ($r = 0$, no spatial extent for propagation), leaving the bimetric interaction ($k^2 = 16$ field coupling channels, the complete 4×4 interaction between two tetrads recognized in the Hassan-Rosen formulation [10]) plus 1 for the non-degenerate coupling vertex itself (the Field Origin, where $\det(e) \neq 0$). The numerator counts the coupling *capacity*: how many field coupling channels are available at the point where the two metrics interact.

The denominator, 13, is the polynomial evaluated in TetB ($s = -1$). Its algebraic structure is: $13 = kd + 1 = 3(3{+}1) + 1$, where $d = 3$ is the spatial dimension. Each of the $k = 4$ face channels projects onto $d = 3$ spatial dimensions, giving $kd = 12$ spatial components (the ADM decomposition [15] that relates the full spacetime interaction to its spatial content), plus 1 for non-degeneracy (the spatial frame must span all three dimensions). The denominator counts the spatial *structure*: what carries the field coupling through space.

The ratio is therefore:

$$\frac{17}{13} = \frac{k^2 + 1}{kd + 1} = \frac{(3+1)^2 + 1}{3(3+1) + 1} = \frac{\text{coupling capacity}}{\text{spatial structure}} \tag{15}$$

This is the structural analog of the 3+1 decomposition. The numerator sees the full bimetric interaction (k^2: both tetrads coupling through the Field Origin, the third component of the polynomial). The denominator sees the spatial projection (kd: the $k = 4$ sub-regions of one coupled face projected onto $d = 3$ spatial dimensions). Neither is adjustable; both follow from $k = (3{+}1)$ and $d = 3$.

The companion paper [1] derived these same counts independently from component analysis of the Hassan-Rosen bimetric action. The structural states derived here from the polynomial provide a second route to the same numbers. The component-counting route and the polynomial route agree because they describe the same field topology from two perspectives. The formula generalizes to arbitrary spatial dimension d, with $d = 3$ uniquely selected by the dual simplex tiling condition $2(d + 1) = 2^d$ (Section IV).

Electron spin from the same topology

The polynomial determines G. The companion paper [1] showed that the same field topology determines electron spin through the face ratio of the stellated octahedron. The structural

states provide an independent route to the same result.

Route 1: The face ratio. Section III D established that the stellated octahedron has 8 faces: 2 TIME faces pointing inward toward the Field Origin, 6 SPATIAL faces pointing outward. Each tetrahedron carries quantized angular momentum $\hbar$ (one closed circuit through 4 faces); the stellated octahedron carries total angular momentum $2\hbar$. Only the 2 TIME faces point toward the vortex eye; the 6 SPATIAL faces carry orbital angular momentum that does not pass through the Field Origin. The observable intrinsic spin is:

$$S = \frac{N_{\text{TIME}}}{N_{\text{total}}} \times L_{\text{total}} = \frac{2}{8} \times 2\hbar = \frac{\hbar}{2} \tag{16}$$

This is the vortex aperture argument: the Field Origin admits angular momentum only through the 2 faces that point toward it.

Route 2: The structural state differences. The structural states reveal a second, independent path. In TetA ($s = +1$), the full hierarchy has 21 field coupling levels. At the Field Origin ($s = 0$), 17 remain. The $21 - 17 = 4$ that vanish are the face channels that require spatial extent ($r > 0$) to propagate. These 4 channels exist only within TetA's (3+1) spacetime sector. Similarly, $17 - 13 = 4$ channels belong to TetB's sector: in TetB ($s = -1$), the opposite orientation inverts the face contribution, removing another 4 channels.

The differences are:

$$21 - 17 = 4 = k = (3{+}1) \quad \text{(face channels in our spacetime sector)}$$
$$17 - 13 = 4 = k = (3{+}1) \quad \text{(face channels in the opposite sector)}$$

The total face contribution across both sectors is $4 + 4 = 8 = 2k$. An observer in our $(3 + 1)$ spacetime has access to only one sector's face channels: 4 of the 8. With quantized angular momentum $\hbar$ per closed circuit through k faces:

$$S = \frac{k}{2k} \times \hbar = \frac{4}{8} \times \hbar = \frac{\hbar}{2} \tag{17}$$

Two independent readings of the same field topology, one geometric and one algebraic, both yield $\hbar/2$. Route 1 counts faces (2 of 8 point inward) with total angular momentum $2\hbar$. Route 2 counts coupling channels from the polynomial (4 of 8 are in our spacetime) with angular momentum $\hbar$ per sector. The intermediate numbers differ (2/8 versus 4/8,

$2\hbar$ versus $\hbar$), but both arrive at $\hbar/2$. Their agreement from independent starting points is a self-consistency check on the topology. The $k = (3{+}1) = 4$ that appears in spin is the same k that appears in the polynomial, in G's prefactor, and in the structural states. The $1/2$ follows from the bimetric structure: an observer occupies one of two coupled spacetime sectors.

One polynomial, three structural states, the gravitational constant, electron spin, and a unique spatial dimension. With the foundation established, we can now derive the equations.

IV. THE (3+1) EQUATIONS

This section presents the four equations that emerge from the $(3 + 1)^2$ topology. We begin at the center: G from tetrad topology, which encodes the $(3 + 1)$ structure in every term. The "1" that appears four times is the Field Origin, where $t = 0$. From this center, three fundamental relationships radiate outward: the inverse square law (connecting Newton and Coulomb), mass-energy equivalence (explaining why c appears squared), and the force hierarchy (resolving the 10^{-43} ratio).

A. G from Tetrad Topology

The equation for G encodes the $(3 + 1)$ structure in every term. The companion paper [1] derived the gravitational constant; here we make the underlying geometry visible. The equation contains a deep insight: the "1" that appears four times is the Field Origin, where $t = 0$.

Where time vanishes and originates

Section III C established that $\tau_{\text{prop}} = r/c = 0$ at the Field Origin, and that the origin is not merely where time vanishes but the source from which time radiates outward. The zero *is* the source.

The equation for G

The gravitational constant, derived in the companion paper [1], can be written with the $(3+1)$ structure visible in every component:

Result 1: G from Tetrad Topology

$$\boldsymbol{G = \frac{(3+1)^2+1}{3(3+1)+1} \times \alpha^{1+(3+1)+(3+1)^2} \times \frac{\hbar c}{m_e^2}} \tag{18}$$

Predicted: $\mathbf{6.666 \times 10^{-11}}$ m^3/(kg·s^2) Measured: 6.674×10^{-11} m^3/(kg·s^2)

Agreement: 0.12%

The 0.12% deviation merits comment. The gravitational constant is the least precisely measured fundamental constant, with inter-experiment scatter of ∼0.05% and CODATA relative uncertainty of 22 ppm. The predicted deviation exceeds both, suggesting either (a) a higher-order correction the topology does not yet capture, analogous to QED corrections to the electron $g-2$, or (b) systematic effects in measurements that future experiments may resolve. This is a falsifiable prediction: if precision measurements converge on a value outside the ∼0.1% window, the topological formula would require revision.

Sensitivity of the exponent. The exponent $1+(3+1)+(3+1)^2 = 21$ is not one of many workable values. Each unit change multiplies the prediction by $1/\alpha \approx 137$:

Exponent	Predicted G	Deviation from measurement
20	9.1×10^{-9}	×137 too large
21	$\mathbf{6.67 \times 10^{-11}}$	**0.12%**
22	4.9×10^{-13}	×137 too small

Adjacent integers miss by over two orders of magnitude. The agreement at 21 is not insensitive to the exponent; it is sharply selective. The value $21 = 1+4+16$ is the only integer in the vicinity that produces a physically reasonable gravitational constant.

The expanded form reveals the underlying geometry; every factor traces to the $(3 + 1)$ structure of spacetime. In fact, the entire equation is determined by a single integer, the spatial dimension d:

$$G = \frac{(d+1)^2 + 1}{3(d+1) + 1} \times \alpha^{1+(d+1)+(d+1)^2} \times \frac{\hbar c}{m_e^2} \tag{19}$$

Setting $d = 3$ produces the prefactor $17/13$, the exponent 21, and the observed gravitational constant. No other value of d yields a universe with both stable orbits and stable atoms (Section IV B).

The four "1"s

The equation for G contains four distinct appearances of "1," each representing the Field Origin from a different perspective:

Location	**Expression**	**Value**	**Physical meaning**
Exponent start	$\mathbf{1} + 4 + 16$	21	The Field Origin ($t = 0$), source of hierarchy
Spacetime structure	$3 + \mathbf{1}$	4	3 space + 1 time radiating from origin
Numerator	$16 + \mathbf{1}$	17	Bimetric channels + origin as coupling interface
Denominator	$12 + \mathbf{1}$	13	Spatial couplings + origin as structural base

TABLE IV. The four appearances of "1" in the equation for G.

Every "1" is the same geometric object viewed from a different angle: the Field Origin where $t = 0$, where the two tetrahedra couple, where gravity's hierarchy begins.

The hierarchy as narrative

The exponent $1 + (3 + 1) + (3 + 1)^2 = 21$ is not merely arithmetic. It tells a story:

1. **The origin (1):** Start at the Field Origin, where $t = 0$. This is the source.
2. **One coupled face (3+1):** From the origin, the field propagates through one coupled face with $3 + 1 = 4$ sub-regions (three spatial, one coupling).

3. **Bimetric coupling $(3+1)^2$:** Both tetrahedra interact. The full $(3+1)\times(3+1) = 16$-dimensional coupling engages.

Gravity couples through all 21 levels. Electromagnetism couples only at level 1. The force hierarchy (Section IV D) follows directly from this narrative.

How the equation encodes all four results

The equation for G is one geometric structure with four readable consequences:

Dimensionality → inverse square law. The $(3 + 1)$ in the prefactor reflects four faces per tetrahedron, the minimal simplex in $d = 3$ dimensions. The same three-dimensionality produces $1/r^2$ through field spreading (Section IV B).

Bimetric structure → c^2. The scale $\hbar c/m_e^2$ contains c in the numerator. The bimetric structure, requiring two tetrahedra with closure speed c each, explains why mass-energy conversion involves $c \times c = c^2$ (Section IV C).

Coupling levels → force hierarchy. The exponent 21 encodes the full hierarchy. Gravity traverses all levels; electromagnetism uses 1. The ratio $\alpha^{20} \approx 10^{-43}$ follows (Section IV D).

Complete equation → G. Combining the $(3 + 1)$ prefactor, coupling (α^{21}), and scale $(\hbar c/m_e^2)$ yields G to 0.12% accuracy.

The origin connects everything

The Field Origin is not incidental. It is the reason gravity exists:

- Without the origin: no $t = 0$ point, no source for time
- Without the coupling point: no bimetric interaction
- Without the "1" in the exponent: no hierarchy

The $(3 + 1)^2$ structure is self-consistent. The number $(3 + 1)$ appears in the dimensions, in the faces, in the prefactor, and in the exponent. The "1" appears as time, as origin, as

coupling interface, and as structural foundation. This result does not merely calculate G; it encodes the geometry of spacetime itself.

The three sections that follow unpack the consequences that radiate from this center.

B. The Inverse Square Law

Newton's gravitational force and Coulomb's electrostatic force share an identical spatial form: both decrease as $1/r^2$. This coincidence has persisted for centuries without geometric explanation. Why do two apparently different forces follow the same distance dependence?

The $(3+1)^2$ topology provides the answer through dimensional geometry. The tetrahedron is the minimal solid in three dimensions: it has $d+1=4$ vertices and faces for $d=3$ spatial dimensions. This relationship between the simplex and dimensionality is fixed by geometry.

Why does space have three dimensions? The answer is physical stability. In $d=2$ dimensions, the gravitational force law $F \propto 1/r$ cannot produce stable closed orbits (Bertrand's theorem [24] permits closed orbits only for $1/r^2$ and r^2 potentials). In $d \geq 4$ dimensions, the force law $F \propto 1/r^{d-1}$ is too steep: the hydrogen atom has no bound states because the centrifugal barrier cannot prevent collapse. Only $d=3$ permits both stable orbits and stable matter. Given $d=3$, the simplex is the tetrahedron with 4 faces. Given the tetrahedron, the $(3+1)^2$ topology follows from the stability requirements of Section III A.

In three-dimensional space, any field emanating from a point source spreads over a spherical surface. The area of that surface at distance r is $4\pi r^2$. If field strength is conserved (flux through any enclosing surface is constant), then intensity at distance r must decrease as the area increases: intensity $\propto 1/r^2$.

The inverse square law is not a separate physical principle. It is the geometry of three-dimensional space. The same $d=3$ that requires the tetrahedron also produces $1/r^2$.

Result 2: The Inverse Square Law

$$\boldsymbol{F \propto \frac{1}{r^2}} \quad \text{(for both gravity and electromagnetism)} \tag{20}$$

Field spreading in three-dimensional space produces $1/r^2$ universally. The tetrahedron's four faces reflect the same $d = 3$.

This result connects Newton and Coulomb. Both wrote:

$$F_{\text{Newton}} = G\frac{m_1 m_2}{r^2} \tag{21}$$

$$F_{\text{Coulomb}} = k\frac{q_1 q_2}{r^2} \tag{22}$$

The $1/r^2$ dependence is identical. Only the prefactors (G versus k) and the sources (mass versus charge) differ. Electromagnetism and gravity are geometric siblings: both obey $1/r^2$ because both propagate through three-dimensional space from the same tetrahedral coupling structure, differing only in coupling depth (α^1 versus α^{21}). The inverse square law is not two separate empirical discoveries; it is one geometric consequence of three-dimensional space.

The companion paper derived the gravitational constant formula [1], where the prefactor $[(3+1)^2+1]/[3(3+1)+1]$ counts coupling channels over structural elements in the dual tetrahedral geometry. The $1/r^2$ spatial dependence derived here is complementary: the prefactor determines the *strength*, while three-dimensional spreading determines the *form*. Together they explain Newton's complete equation.

C. Mass-Energy Equivalence

Section III E established why $E = mc^2$ takes this form: mass is energy bound at the Field Origin (the "1"), and conversion engages both tetrahedra, each with closure speed c. The "squared" in c^2 has the same origin as the "squared" in $(3+1)^2$. Here we present this as a formal result and trace its connection to G from tetrad topology.

Result 3: Mass-Energy Equivalence

$$\boldsymbol{E = m \times c_{\text{TetA}} \times c_{\text{TetB}} = m \times c \times c = mc^2} \tag{23}$$

The c^2 factor is the product of two closure speeds, one from each tetrahedron.

The speed c appears in both Einstein's mass-energy relation and Maxwell's electromagnetic equations. In the $(3+1)^2$ framework, this is not coincidence: both emerge from the closure speed of the tetrahedral structure. Any field propagating through the $(3+1)$ topology must travel at c, whether it carries gravitational, electromagnetic, or any other interaction.

This result connects directly to the topological derivation of G. The formula contains c in the numerator (via $\hbar c$) and m_e^2 in the denominator. Both appearances trace to the same bimetric source:

- The c in $\hbar c$ reflects one closure speed (field propagation)
- The m_e^2 in the denominator reflects energy bound at the Field Origin, squared because both tetrahedra participate

Einstein's c^2 and the $\hbar c/m_e^2$ scale in G share the same geometric origin: the two-tetrahedron structure requires two factors wherever mass or energy crosses between the bimetric sectors. The closure speed also explains why c is the universal speed limit: in the $(3+1)$ structure, every object moves through spacetime at the invariant speed c, partitioned as $c^2 = v_{\text{space}}^2 + v_{\text{time}}^2$. Moving faster through space leaves less budget for moving through time, which is why clocks slow at high velocities and stop entirely at $v = c$.

D. The Force Hierarchy

Section IV B established that Newton's gravity and Coulomb's electromagnetism share the same spatial form: $F \propto 1/r^2$. Yet gravity is approximately 10^{43} times weaker. If the

geometry is identical, why do the strengths differ by such an enormous factor?

The answer lies not in spatial form but in coupling depth, the number of structural levels a force must traverse to complete its interaction. Think of coupling depth as floors in a building: a signal that must pass through more floors arrives weaker at the other end. The $(3+1)^2$ topology contains a hierarchy of coupling channels, established in Section III E: the Field Origin (1 channel), one coupled face (4 channels), and the full bimetric interaction (16 channels), totaling 21. Different forces engage different numbers of these channels.

The difference between electromagnetism and gravity lies in what they couple. Electromagnetic interactions involve charge, which exists within a single spacetime. A photon couples to a charged particle at one point in one tetrahedron's structure. This physical picture suggests that electromagnetism engages only the origin channel, giving coupling strength proportional to α^1. This assignment is motivated by the physical distinction between local (EM) and bimetric (gravitational) coupling; the numerical match to the observed hierarchy provides empirical support.

Gravitational interactions involve mass, and mass is energy bound at the Field Origin, where both tetrahedra meet. Gravity must therefore couple through the complete bimetric structure: the origin (1 channel), one coupled face (4 channels), and the full $(3+1)\times(3+1)$ cross-coupling between both tetrahedra (16 channels). The coupling strength is proportional to α^{21}.

Result 4: The Force Hierarchy

$$\frac{\text{Gravity}}{\text{EM}} \sim \frac{\boldsymbol{\alpha^{21}}}{\boldsymbol{\alpha^{1}}} = \boldsymbol{\alpha^{20}} \approx \left(\frac{\mathbf{1}}{\mathbf{137}}\right)^{\mathbf{20}} \approx \mathbf{10^{-43}} \tag{24}$$

Same spatial form ($1/r^2$), different coupling depth (21 vs. 1).

The hierarchy dissolves. Gravity is not mysteriously weak. It is naturally attenuated because it must traverse the complete geometric hierarchy: 21 coupling channels versus electromag-

netism's single channel. The 20-order difference in coupling exponents produces the 43-order difference in observed strength. No fine-tuning is required; the ratio follows from counting levels in the $(3+1)^2$ topology.

This result completes the connection between Newton and Maxwell. Both forces follow $1/r^2$ because both propagate through three-dimensional space (Section IV B). They differ in strength because electromagnetism couples locally while gravity engages the full bimetric hierarchy. The spatial form is shared; the coupling depth is distinct.

V. QUANTUM CONSISTENCY OF THE $(3+1)^2$ TOPOLOGY

Section IV derived quantitative predictions from the $(3+1)^2$ topology: G to 0.12%, the force hierarchy from coupling depth, and the geometric origins of $1/r^2$ and $E = mc^2$. A natural question follows: is the same topology consistent with quantum mechanics?

The $(3+1)^2$ structure operates at the scale of the electron radius, deep within the quantum regime. A geometry that determines G at this scale would be expected to constrain quantum behavior at the same scale. This section examines three aspects of quantum physics through the topological lens. The first, energy quantization, provides a conceptual mechanism through the Field Origin as a boundary condition. The second and third, wave-particle duality and the energy-momentum relation, demonstrate that the topology is consistent with established relativistic quantum mechanics. These vary in rigor; the section is transparent about what is derived, what is interpreted, and what is inherited from standard physics.

A. Energy Quantization

Classical physics assumed energy was continuous: any amount could exist. Experiments revealed otherwise. Heated objects emit light in discrete colors. Atoms absorb only specific frequencies. Energy comes in packets.

Planck's relation captures this:

$$E = hf \tag{25}$$

where h is Planck's constant and f is frequency. But this equation was introduced empiri-

cally. Planck called it "an act of desperation" [26]. The question remained: why is energy quantized?

The $(3+1)^2$ framework provides a conceptual answer through resonance. The Field Origin acts as a topological boundary condition. Consider a guitar string: fixed endpoints permit only standing waves with specific wavelengths. Similarly, the Field Origin's structure selects which field configurations can stably exist. Configurations that do not "fit" the boundary cannot persist as stable resonances. This does not derive the value of h from topology, but it explains *why* quantization occurs: discrete modes are the only stable solutions.

Just as a guitar string's length determines which harmonics can sound, the Field Origin's topology determines which frequencies are allowed. The discrete frequencies that satisfy this boundary condition correspond to quantized energies through $E = hf$. Intermediate configurations, those that violate the boundary condition, cannot exist as stable modes. The Field Origin is the universal boundary that selects allowed resonances.

Result 5: Energy Quantization

$$\boldsymbol{E = hf} \tag{26}$$

The Field Origin selects discrete resonance modes. Energy is quantized because only standing waves that fit the origin can exist.

Planck's constant $h = 6.626 \times 10^{-34}$ J·s is small because the Field Origin is at the smallest scale. It represents the "grain size" of resonance, the minimum action required for one complete cycle at the coupling point. A wave oscillating f times per second completes f cycles in one second. Each cycle requires action h to execute. The energy of such a wave is therefore $E = hf$: action per cycle times cycles per second.

Quantization is not strange. It is what resonance does. Guitar strings have harmonic series. Organ pipes produce specific notes. Atoms have discrete energy levels. The Field Origin is

the universal boundary that selects allowed modes.

B. Wave-Particle Duality

Light sometimes behaves as a wave (interference, diffraction) and sometimes as a particle (photoelectric effect, Compton scattering). Matter exhibits the same duality: electrons diffract through crystals like waves, yet they register as discrete hits on detectors. De Broglie proposed that every particle has an associated wavelength:

$$\lambda = \frac{h}{p} \tag{27}$$

where p is momentum. This relation bridged quantum and classical physics. But it appeared as a postulate, unexplained.

The $(3{+}1)^2$ framework shows that wave-particle duality is a consequence, not an assumption. From the preceding results, energy is quantized: $E = hf$. For light, we also know that photons are massless and travel at speed c. A photon's energy is entirely kinetic, carried by its momentum: $E = pc$. (This is the massless limit of the general energy-momentum relation derived in Section V C.)

Combining quantization with the photon relation:

$$E = hf = pc \tag{28}$$

$$h \cdot \frac{c}{\lambda} = pc \tag{29}$$

$$\lambda = \frac{h}{p} \tag{30}$$

De Broglie's insight was that this relation applies not just to photons but to all particles. Once the Field Origin selects discrete energy modes and the bimetric structure sets the energy-momentum relation, wave-particle duality emerges automatically. Every particle with momentum p has an associated wavelength $\lambda = h/p$.

Consistency Check: Wave-Particle Duality

$$\lambda = \frac{h}{p} \tag{31}$$

De Broglie's relation follows algebraically from quantization ($E = hf$) combined with the relativistic energy-momentum relation. The topology contributes indirectly through the boundary conditions that produce quantization; the algebraic step is standard physics.

The physical picture: a "particle" is a resonance at a Field Origin. A "wave" is how that resonance propagates through $(3 + 1)$ space. They are not contradictory. They are two aspects of the same phenomenon. Detection (particle behavior) occurs at the Field Origin. Propagation (wave behavior) occurs through the surrounding spacetime. The duality reflects the structure: localized origin, extended field.

C. The Energy-Momentum Relation

Section IV C derived $E = mc^2$ from the bimetric structure: two tetrahedra, each with closure speed c, producing c^2. But $E = mc^2$ applies only to objects at rest. Moving objects have both rest mass energy and kinetic energy. What is the complete relationship?

Einstein's energy-momentum relation answers:

$$E^2 = (pc)^2 + (m_0c^2)^2 \tag{32}$$

This equation has the structure of the Pythagorean theorem in energy-momentum space. Rest mass energy m_0c^2 and momentum contribution pc combine as orthogonal components. Total energy E is the hypotenuse.

Why does energy-momentum follow Pythagorean geometry? The $(3+1)^2$ framework provides the answer: the invariant interval.

Consistency Check: The Energy-Momentum Relation

$$E^2 = (pc)^2 + (m_0c^2)^2 \tag{33}$$

The energy-momentum relation follows from the invariance of the $(3+1)$ spacetime interval. The topology provides the $(3+1)$ structure; the Pythagorean form follows from Lorentz invariance, which is standard special relativity.

The derivation follows from the spacetime interval. In the $(3+1)$ structure, the interval $ds^2 = c^2dt^2 - dx^2$ is invariant. All observers agree on ds, regardless of relative motion. This invariance propagates to energy-momentum space: the quantity $E^2 - (pc)^2$ must be invariant.

For an object at rest in some frame, $p = 0$ and $E = m_0c^2$. Therefore:

$$E^2 - (pc)^2 = (m_0c^2)^2 - 0 = (m_0c^2)^2 \tag{34}$$

This must hold in all frames. Rearranging:

$$E^2 = (pc)^2 + (m_0c^2)^2 \tag{35}$$

The Pythagorean form is not assumed. It follows from demanding that the $(3+1)$ interval remain invariant across reference frames. The structure of spacetime determines the structure of energy-momentum.

Two limiting cases confirm the result. For a particle at rest ($p = 0$): $E = m_0c^2$, recovering Einstein's equation. For a massless particle ($m_0 = 0$): $E = pc$, the photon relation. Both follow from the same geometric principle.

VI. THE FINE STRUCTURE CONSTANT

The seven equations derived in Sections IV and V all contain α, but treat it as an empirical input. This section derives α itself. The derivation proceeds in three steps: first, why α is a topological quantity shared by both forces; second, the integer $1/\alpha = 137$ from counting the total interaction states of the stellated octahedron; third, independent confirmation through measurable physical rates.

A. Two Scales, One Topology

Why does the fine-structure constant, a measure of electromagnetic coupling strength, appear in a gravitational formula? The answer, established by the convergence in Section II D, is that α is not exclusively electromagnetic. It is a property of the topology itself.

Section II D demonstrated that two independent routes converge on the same characteristic scale. The electromagnetic route defines the classical electron radius, the distance at which electromagnetic self-energy balances rest mass energy:

$$r_e = \frac{e^2}{4\pi\varepsilon_0 m_e c^2} = 2.818 \times 10^{-15} \text{ m} \tag{36}$$

This uses only electromagnetism and special relativity. No gravity enters. The gravitational route, using cosmological and topological inputs with no electromagnetic data, independently finds $r_e = 2.82 \times 10^{-15}$ m (Table II). The match is 0.21%.

Two forces, two independent calculations, one scale. This convergence has a structural explanation. Consider the speed of sound in a crystal. It depends on two things: the lattice spacing and the bond strength. Thermal vibrations and acoustic waves both propagate through the same lattice, and both depend on the same structural parameters. They are not "borrowing" from each other; they share the same physical structure. Similarly, electromagnetism and gravity share the same topological scale r_e because both operate on the same underlying field structure: the stellated octahedron with its $k = 4$ coupling channels.

The Compton wavelength provides a second characteristic scale:

$$\lambda_C = \frac{h}{m_e c} = 2.426 \times 10^{-12} \text{ m} \tag{37}$$

This is the wavelength of a photon whose energy equals the electron's rest mass. It sets the quantum mechanical scale of the electron.

The fine-structure constant is the ratio of these two topological scales:

$$\alpha = \frac{r_e}{\bar{\lambda}_C} \qquad \text{where } \bar{\lambda}_C = \frac{\hbar}{m_e c} = \frac{\lambda_C}{2\pi} \tag{38}$$

Alpha measures how the field's characteristic scale r_e compares to the quantum scale $\bar{\lambda}_C$. It is the topology's characteristic ratio. Electromagnetism probes this ratio at one coupling vertex (the interaction point where a charge couples to a field), contributing one factor of α. Gravity traverses the entire coupling hierarchy of the stellated octahedron, contributing α^{21}. Both forces use the same α because both originate from the same topological structure.

B. The Breadth of the Hierarchy

Having established that α is a topological ratio, we now ask: what determines its value? The answer lies in the same stellated octahedron whose depth gave the exponent in G.

Section III E counted along one path through the coupling hierarchy: $1 + 4 + 16 = 21$. This depth gives the exponent in G. But a path through a structure is not the structure itself. A tree has a height and a canopy. The height tells you how far you must climb from root to tip. The canopy tells you how many leaves the tree contains. Both numbers describe the same tree, but they answer different questions. The height of the coupling hierarchy is 21. What is its canopy?

Counting the breadth

The depth traced one path through one coupled face (Figure 4). The breadth counts every interaction state across the complete stellated octahedron.

Start at the center.

The Field Origin contributes one state. This is the same "1" that appears at the base of the coupling hierarchy and as the first term of the polynomial $f(k) = 1 + k + k^2$ (Section III E). The non-linear coupling boundary at $r = 0$ has no internal decomposition; it is the single

topological fact that the two sectors couple. Its contribution to the breadth is 1, exactly as its contribution to the depth is 1.

The vertices. The depth count used $k = 4$, the four sub-regions of one coupled face (Figure 4: three spatial sub-regions plus one coupling sub-region), because it traced a single path through one sector. The breadth sees the entire structure. Each tetrahedron has $k = 4$ vertices; the stellated octahedron contains two tetrahedra, giving $2k = 8$ vertices in total. This factor of 2 is the same one that produces electron spin $\hbar/2$ (Section III F): our $(3 + 1)$ spacetime observes one tetrahedron's contribution ($k = 4$), but the full structure contains both ($2k = 8$).

The modes per vertex. At each vertex, the full bimetric interaction space is accessible: $k^2 = 16$ interaction channels (the complete 4×4 coupling between the two tetrads, the third term of the polynomial), plus one non-degeneracy condition ($\det(e) \neq 0$) ensuring the local frame is well-defined. That gives $k^2 + 1 = 17$ interaction modes per vertex, the same 17 that appears as the numerator of the prefactor 17/13 (Section III F).

The product. Eight vertices, seventeen modes each. Each vertex independently accesses all 17 modes, just as each floor of a building independently contains all of its rooms. The product is $8 \times 17 = 136$. Adding the Field Origin:

$$N = 1 + 2k(k^2 + 1) = 1 + 8 \times 17 = 137$$

One center, plus eight vertices times seventeen modes each. The integer 137 is the total interaction state count of the stellated octahedron. Every number in this count was derived independently in Section III: the Field Origin (1), the vertex structure ($2k = 8$), and the interaction modes ($k^2 + 1 = 17$).

The topological correction

The integer 137 counts discrete interaction states. But these states do not float in isolation; they live on a geometric scaffold.

The stellated octahedron has 28 geometric elements: 8 vertices, 12 edges, and 8 faces (Section III B). These 28 elements are the scaffolding that connects the 137 discrete states to one another: the vertices where states reside, the edges along which they communicate, and

the faces through which they couple.

The f-vector sum $V + E + F = 28$ decomposes entirely in terms of $k = (3+1) = 4$ and $d = 3$:

- $2k = 8$ vertices (two tetrahedra, each with k vertices)
- $dk = 12$ edges (two tetrahedra, each with $k(k-1)/2 = 6$ edges, and $k - 1 = d$)
- $2k = 8$ faces (two tetrahedra, each with k triangular faces)

The total is $2k + dk + 2k = k(k + d) = 4 \times 7 = 28$. The same k and d that determine the depth (21), the prefactor (17/13), and every integer in G also determine the geometric complexity of the scaffold.

This is standard topological index structure. In crystallography, atoms at different positions in a unit cell contribute fractional counts based on the geometric elements they occupy: a corner atom shared among eight cells contributes 1/8 per cell, an edge atom shared among four contributes 1/4, a face atom shared between two contributes 1/2. The total atom count per cell is the sum of these fractional geometric contributions. The same principle applies here. The stellated octahedron is one complete topological structure with 28 geometric elements. One indivisible structure contributes:

$$\frac{1}{k(k+d)} = \frac{1}{28}$$

The integer 137 counts discrete states (like the number of atoms). The fraction 1/28 is the contribution of the continuous geometric scaffold on which those states live (like the fractional position within the unit cell). The complete topological index is:

$$\frac{1}{\alpha} = 1 + 2k(k^2 + 1) + \frac{1}{k(k+d)} = 137 + \frac{1}{28} = 137.0357\ldots \tag{39}$$

The CODATA 2022 value is $1/\alpha = 137.035\,999\,177$. The agreement is **0.00021%**, matching to five significant figures.

Depth and breadth: the same topology read two ways

The depth column is the exponent in G; the breadth column counts the total interaction states.

Structure	Depth (one path)	Breadth (all states)
Field Origin	1	1
One coupled face	$k = 4$	$2k \cdot 1 = 8$
Bimetric interaction	$k^2 = 16$	$2k \cdot k^2 = 128$
Integer total	**21**	**137**
Geometric scaffold	—	$+\,1/[k(k+d)] = 1/28$

The depth column sums the three polynomial components along one path: $1 + 4 + 16 = 21$. The breadth column counts all states across the full stellated octahedron. Each coupled face has $k = 4$ sub-regions (Figure 4); two tetrahedra give $2k = 8$ vertices total. Each vertex accesses $k^2 + 1 = 17$ modes, so the breadth decomposes as: 1 (Field Origin) + 8×1 (vertex contributions) + 8×16 (bimetric contributions) = 137. The depth gives α^{21} in G. The breadth gives $1/\alpha = 137.036$. The α equation is the G equation read sideways: not how deep the hierarchy runs, but how many states it contains.

Every integer in Eq. (39) uses only $k = (3{+}1) = 4$ and $d = 3$. No additional quantities are introduced. The same topology that determines G determines α.

C. Confirmation from Physical Rates

The breadth of the hierarchy predicts $1/\alpha = 137.036$. This prediction can be confirmed independently through measurable physical rates, without counting states.

Section II B established that field closure requires all $k = 4$ face channels to settle into a consistent state. The speed of this reconciliation is c (Lorentz invariance admits no other fundamental speed), and the distance to be traversed is r_e. The rate at which the field completes one full consistency cycle is:

$$f_{\text{closure}} = \frac{c}{r_e} \tag{40}$$

This is the field closure rate: how fast the topology's internal consistency process operates across the four sub-regions of one coupled face.

The electron also has an internal clock. Quantum mechanics relates mass to frequency through $E = mc^2$ and $E = hf$. The electron's intrinsic oscillation rate is:

$$f_{\text{Compton}} = \frac{m_e c^2}{h} \tag{41}$$

This is the Compton frequency, the tick rate of the electron's quantum phase.

Two rates, both physical, both measurable. One describes how fast the field maintains consistency across its coupling channels. The other describes how fast the particle's quantum state evolves. Their ratio reveals the fine-structure constant:

$$\frac{f_{\text{closure}}}{f_{\text{Compton}}} = \frac{c/r_e}{m_e c^2/h} = \frac{h}{m_e c \cdot r_e} = \frac{\lambda_C}{r_e} = \frac{2\pi}{\alpha} \tag{42}$$

Therefore:

$$\frac{1}{\alpha} = \frac{1}{2\pi} \cdot \frac{f_{\text{closure}}}{f_{\text{Compton}}} = \frac{\lambda_C}{2\pi r_e} = 137.036 \tag{43}$$

The rate ratio independently confirms the topological count of Section VI B.

The connection between these two routes is the field closure condition. If the topology contains N independent interaction states, the field must cycle through all of them once per Compton period for the configuration to remain stable. This is the same quantization principle that Bohr applied to atomic orbits: only configurations where every state completes a full phase cycle (2π) per quantum period are self-consistent. The closure rate must therefore satisfy:

$$f_{\text{closure}} = N \times 2\pi \times f_{\text{Compton}}$$

Comparing with $f_{\text{closure}}/f_{\text{Compton}} = 2\pi/\alpha$ gives $2\pi/\alpha = 2\pi \times N$, and therefore:

$$\frac{1}{\alpha} = N = 1 + 2k(k^2 + 1) = 137 \tag{44}$$

The state count equals the inverse fine-structure constant because the field must cycle through all N states per quantum tick to maintain closure. The breadth of the hierarchy (Section VI B) counts the states; the rate ratio measures the same quantity from physical observables.

Result 6: The Fine Structure Constant

The fine structure constant is the ratio of two physical rates, per angular cycle:

$$\frac{1}{\alpha} = \frac{1}{2\pi} \cdot \frac{c/r_e}{m_e c^2/h} = 137.036 \tag{45}$$

The integer 137 counts the total interaction states of the stellated octahedron; the fractional correction arises from its 28-element geometric scaffold.

The number $1/\alpha = 137$ counts how many times the field structure updates its internal consistency for each cycle of the electron's quantum phase. Alpha is the per-vertex coupling strength at the Field Origin, set by the topology's own characteristic scales. Electromagnetism engages one coupling vertex, contributing α^1. Gravity traverses all 21 coupling levels of the hierarchy (Section III E), contributing α^{21}. The force ratio $\alpha^{21}/\alpha^1 = \alpha^{20} \approx 10^{-43}$ (Section IV D) follows directly: gravity is weaker because it couples deeper, not because it is fundamentally different.

D. What This Means

The fine structure constant has puzzled physicists since Sommerfeld introduced it in 1916. Feynman called it "one of the greatest damn mysteries of physics" [25]. Pauli reportedly said that when he died, his first question to the Devil would be: why 137?

The answer is topological. The integer 137 counts the total interaction states of the stellated octahedron: 1 at the Field Origin, plus $2k = 8$ vertices (two tetrahedra, each with $k = 4$) each accessing $k^2 + 1 = 17$ bimetric modes. Every number traces to the polynomial $f(k) = 1 + k + k^2$ and its three geometric components: the Field Origin (1), the coupled face ($k = 4$ sub-regions), and the bimetric interaction ($k^2 = 16$ channels). The fractional correction 1/28 follows from the 28 geometric elements of the stellated octahedron, a count that decomposes as $k(k + d)$ using the same building blocks. No new assumptions or parameters are introduced.

Once the topology determines the integers in G (Section III) and the characteristic scales (Section II D), α follows. The fine-structure constant is not a free parameter alongside G; both are determined by the same $(3+1)^2$ topology, one by reading the coupling hierarchy for its depth, the other for its breadth.

With α derived, a natural question arises: can the electron mass itself be derived? Section VII F demonstrates that it can. Three overdetermined equations from gravity, cosmology, and electromagnetism converge on a unique solution for m_e, removing the last empirical particle-physics input. The topology determines not only the relationships but the scale.

VII. CONCLUSION

A. The Logic Chain

The results derived in this paper follow from an unbroken chain of necessity. Stability requires three spatial dimensions: fewer produce weak long-range forces, more produce unstable atoms. Three dimensions determine the tetrahedron as the minimal simplex. A single tetrahedron is unstable, creating ghost modes. Stability requires two tetrahedra with opposite orientations. The stellated octahedron is the unique configuration satisfying stability, spinor structure, chirality, and ghost-free dynamics.

From this required structure, nine results follow:

The (3+1) Equations (Section IV):

- Coupling channels over structural elements $[(3+1)^2+1]/[3(3+1)+1]$ plus α^{21} yield G to 0.12%
- Three-dimensional spreading yields $1/r^2$
- Two closure speeds yield c^2
- Coupling levels $1+4+16=21$ yield $\alpha^{20} \approx 10^{-43}$

Quantum Consistency (Section V):

- The Field Origin as boundary selects discrete modes: $E = hf$

- Quantization plus relativity yield $\lambda = h/p$
- Interval invariance yields $E^2 = (pc)^2 + (m_0c^2)^2$

The Fine Structure Constant (Section VI):

- The ratio of field closure rate to Compton frequency identifies α as the per-vertex coupling strength
- Breadth counting of the stellated octahedron's interaction states yields $1/\alpha = 137.036$ to 0.00021%

The chain connects Newton, Maxwell, Einstein, Planck, and de Broglie, and resolves the dark matter problem. The topology is not chosen; it is required. Once required, the equations follow.

B. What Has Been Connected

These equations are not merely a list. They resolve questions that stood open for decades or centuries.

Force Laws. Newton published the gravitational inverse square law in 1687. Coulomb established the same law for electrostatics in 1785. For over two centuries, physicists noted the parallel but could not explain it. The $(3+1)^2$ topology provides the answer: both forces spread through three-dimensional space from point sources. The tetrahedron is the minimal simplex in $d = 3$ dimensions; in three dimensions, flux through any enclosing surface dilutes as $1/r^2$. The inverse square law is geometry.

Mass and Energy. Einstein's $E = mc^2$ contains a squared speed of light, but why c^2 rather than c or c^3? The bimetric structure answers this: two tetrahedra, each with its own closure speed, both equal to c. The product of two speeds gives c^2. Mass-energy equivalence reflects the dual structure of spacetime.

The Hierarchy. Gravity is 10^{43} times weaker than electromagnetism. This ratio appears arbitrary until the coupling hierarchy is traced: $1+(3+1)+(3+1)^2 = 21$ levels, with gravity coupling at depth 21 and electromagnetism at depth 1. The ratio $\alpha^{21}/\alpha^1 = \alpha^{20} \approx 10^{-43}$. The hierarchy is not arbitrary; it counts coupling levels.

Quantum Foundations. Energy quantization, wave-particle duality, and the energy-momentum relation appear in textbooks as separate postulates. The topology addresses all three. The Field Origin as a boundary condition selects discrete resonant modes. Quantization plus relativity yields the de Broglie relation. Interval invariance yields the relativistic energy-momentum relation. Three quantum results, one geometric origin.

Constants. The gravitational constant G and fine structure constant α have been measured for over a century but never explained. The topology derives both: G from the $(3+1)$ structure and coupling depth (0.12% accuracy), α from the total interaction states of the stellated octahedron (0.00021% accuracy). Constants that appeared fundamental are now consequences of structure.

These connections span three centuries of physics. Newton, Coulomb, Einstein, Planck, de Broglie, and Sommerfeld each contributed pieces. The $(3+1)^2$ topology reveals that the pieces fit together.

C. Comparison with Other Approaches

To appreciate what this framework offers, it helps to compare it with established alternatives. The question is not which approach is "best," but which achieves specific goals.

Approach	G	$1/r^2$	$\hbar/2$	α	**Testable?**
Standard Model	Outside scope	Axiom	Axiom	Input	Yes
String Theory	Landscape	Derived	Partial	Landscape	Not yet
Loop Quantum Gravity	Not derived	Partial	Partial	No	Partial
$(3+1)^2$ Framework	**Predicted (0.12%)**	**Consistent**	**Geometric**	**Derived (0.00021%)**	**Yes**

TABLE V. Comparison of how different frameworks address selected quantities. "Predicted" and "Derived" indicate zero-parameter numerical predictions with stated accuracy. "Consistent" means the topology produces the result but the derivation uses standard physics (Gauss's law in $d = 3$). "Geometric" indicates a novel structural argument (the face ratio determines spin $\hbar/2$ from the stellated octahedron, Section III F). Each framework has different goals; the comparison reflects scope, not quality.

The distinguishing feature of the $(3+1)^2$ framework is its testable numerical predictions: G to 0.12%, $1/\alpha$ to 0.00021%, a_0 to 0.41%, m_e to 0.1%, and H to 0.11%, all with zero adjustable parameters. The quantum results in Section V range from a conceptual mechanism (the Field Origin as boundary condition producing quantization) to consistency checks (wave-particle duality and energy-momentum from standard relativistic physics).

D. Resolution of the Dark Matter Problem

Why does standard cosmology require 85% of gravitational mass to be invisible? The answer lies in a hidden assumption: that spacetime is a structureless continuum through which gravity propagates uniformly at all scales. This assumption, while adequate for solar-system dynamics, fails at galactic scales. The $(3+1)^2$ topology reveals the error and provides the correction.

The hidden assumption

Consider calculating ocean tides while assuming water is a structureless continuum. Such a calculation would fail because water has molecular structure: H_2O molecules arrange in tetrahedral electron geometry, and this structure creates bulk properties (viscosity, surface tension, specific heat) that govern large-scale behavior. Ignoring molecular structure while trying to predict tides would require inventing "dark forces" to explain the discrepancy.

The same error occurs in gravitational physics. Standard cosmology treats spacetime as structureless, but fields couple, and coupling creates topology. Just as water molecules form tetrahedral arrangements, the gravitational field couples through the $(3+1)^2$ dual tetrad structure. This structure creates a bulk property: a characteristic acceleration scale a_0 below which the coupling changes character.

The characteristic scale

The $(3+1)^2$ topology yields a characteristic acceleration:

$$a_0 = \frac{k^2\, G\, m_e}{r_e^2} = \frac{16\, G\, m_e}{r_e^2} \approx 1.2 \times 10^{-10}\ \mathrm{m/s^2} \tag{46}$$

where $k^2 = 16$ counts the coupling channels through the tetrahedral geometry. This derived value matches the empirically observed MOND acceleration scale [21] to 0.41% [2]. The scale a_0 marks a physical boundary: above it, all 16 coupling channels engage and gravity behaves as Newton described. Below it, the field cannot sustain full coupling, and the effective gravitational force must be enhanced to achieve closure.

The correction factor

What happens below a_0? At each point in the gravitational field, the available coupling capacity can be divided into two states: a *resolved* fraction ε that successfully tracks the Newtonian demand, and an *unresolved* fraction $(1-\varepsilon)$ where the coupling has saturated and the field cannot respond further. Think of a highway: at low traffic (high g), all lanes carry vehicles efficiently. At high traffic (low g), some lanes saturate, and the remaining traffic must redistribute.

The equilibrium between these two states follows from detailed balance. Two competing rates govern the partition:

- **Resolution rate:** $W_+ = \kappa \cdot x \cdot (1-\varepsilon)$. The field is driven toward resolution by the gravitational intensity $x = g/a_0$, but limited by the available (unresolved) capacity $(1-\varepsilon)$.
- **Relaxation rate:** $W_- = \kappa \cdot \varepsilon$. Resolved capacity relaxes back at a rate proportional to the resolved fraction ε.

At equilibrium, $W_+ = W_-$:

$$\kappa \cdot x \cdot (1-\varepsilon) = \kappa \cdot \varepsilon \tag{47}$$

Solving for ε:

$$\varepsilon = \frac{x}{1+x} \tag{48}$$

Since the resolved fraction *is* the fraction of Newtonian gravity that the field transmits, $\mu(x) \equiv \varepsilon(x)$:

$$\mu(x) = \frac{x}{1+x} \quad \text{where} \quad x = \frac{g}{a_0} \tag{49}$$

This result is unique within the class of single-channel saturating closures with linear response [4]: the exponent of x in the driving rate is forced to be 1 by the requirement that $\mu/x \to 1$ as $x \to 0$ (the deep-MOND limit observed in galaxy rotation curves). Any other exponent violates this limit. The "1" in the denominator represents the always-present baseline coupling channel, the Field Origin in the hierarchy $1 + 4 + 16 = 21$. Even when all other channels saturate, this single channel persists. Figure 6, top right, shows $\mu(x)$: the red shaded region ($x < 1$) is where spacetime structure enhances gravity; the blue shaded region ($x > 1$) is where Newton applies.

What the correction produces

The correction factor $\mu(x)$ has three consequences, each observable.

First: how gravity weakens with distance changes. At high accelerations ($g \gg a_0$), $\mu \to 1$ and gravity weakens as $1/r^2$, following Newton. At low accelerations ($g \ll a_0$), $\mu \to x$ and the effective gravity becomes:

$$g_{\text{eff}} = \sqrt{g_{\text{Newton}} \times a_0} = \frac{\sqrt{GMa_0}}{r} \propto \frac{1}{r} \tag{50}$$

This transition is a smooth crossover, not an abrupt switch: $\mu(x)$ is analytic everywhere. Figure 6, bottom left, shows how gravity weakens with distance on logarithmic axes: the blue curve tracks Newton's $1/r^2$ at small radii and bends to follow $1/r$ at large radii.

Second: rotation curves are flat. For circular orbits, $v^2/r = g_{\text{eff}}$. In the deep-MOND regime where $g \ll a_0$:

$$v^2 = g_{\text{eff}} \times r = \sqrt{GMa_0} = \text{constant} \tag{51}$$

The rotation velocity is independent of radius: the curve is flat. This is precisely what astronomers observe in every spiral galaxy. In the Newtonian calculation, $v \propto 1/\sqrt{r}$, a declining curve that contradicts observation. Figure 6, top left, shows both: the red dotted line (Newton) declines; the blue curve (topology) rises and flattens, matching galactic data.

Third: the 85% "dark matter" is explained. At any radius, an observer using Newtonian gravity would infer more mass than is visible. The apparent "dark matter" fraction is:

$$f_{\text{dark}} = 1 - \mu(x) = \frac{1}{1+x} = \frac{a_0}{g + a_0} \tag{52}$$

At typical galaxy outskirts where $g \approx 0.1\, a_0$, this gives $f_{\text{dark}} \approx 90\%$. Integrated over a galaxy, the average is approximately 85%, matching observations. Figure 6, bottom right, shows this: the grey shaded area is the mass unaccounted for at each radius when Newtonian gravity ($1/r^2$) is assumed, rising from 18% at 5 kpc to 95% at 50 kpc. The horizontal line at 85% is the cited dark matter fraction. The blue line at zero shows the corrected gravity ($1/r^2 \to 1/r$): no mass is unaccounted for. The entire grey region is a calculation artifact from using the wrong distance dependence.

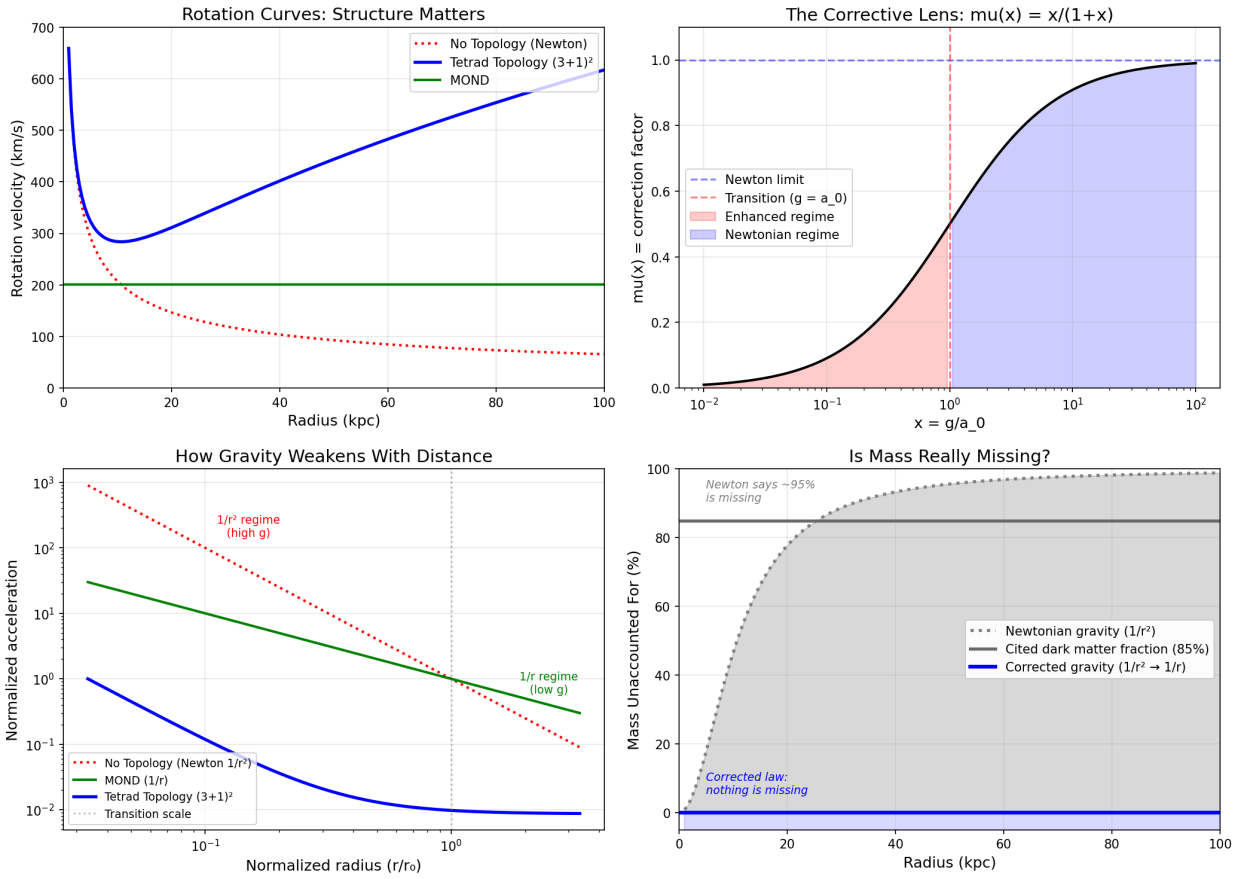


FIG. 6. The dark matter problem as a structural correction, shown for a $10^{11} M_\odot$ galaxy. Each panel corresponds to one consequence of the correction factor $\mu(x) = x/(1+x)$ derived above. **Top left:** Rotation curves (Newton declining in red, topology in blue, deep-MOND asymptote in green). **Top right:** The correction factor itself, showing the enhanced (red) and Newtonian (blue) regimes. **Bottom left:** How gravity weakens with distance: the transition from $1/r^2$ to $1/r$ on logarithmic axes. **Bottom right:** Is mass really missing? The grey curve shows the mass unaccounted for at each radius using Newtonian gravity ($1/r^2$). The horizontal line marks the cited 85% dark matter fraction. The blue line shows the corrected gravity ($1/r^2 \to 1/r$): no mass is unaccounted for. See text for detailed descriptions.

Table VI quantifies the predictions at specific radii.

r (kpc)	g/a_0	$\mu(x)$	v_{Newton} (km/s)	v_{actual} (km/s)	"DM" %
5	4.71	0.83	295 km/s	324 km/s	18%
10	1.18	0.54	208	283	46%
20	0.30	0.23	147	309	77%
30	0.13	0.12	120	354	88%
50	0.05	0.05	93	439	95%

TABLE VI. Rotation curve predictions for a $10^{11}M_\odot$ galaxy at specific radii. As radius increases, g/a_0 decreases and $\mu(x)$ drops, meaning the structural correction grows larger. At 30 kpc, where most rotation curve measurements are made, the framework predicts 88% apparent dark matter, matching observations without invoking invisible mass.

The corrective lens

Standard calculations assume $M_{\text{inferred}} = v^2 r/G$. The correct relationship is:

$$M_{\text{actual}} = M_{\text{inferred}} \times \mu(g/a_0) = M_{\text{inferred}} \times \frac{g}{g + a_0} \tag{53}$$

This is the corrective lens that transforms standard cosmological calculations. Applying it removes the need for dark matter: the "missing" 85% is restored as the correction factor, not as additional mass.

Relationship to MOND and known challenges. The correction factor $\mu(x) = x/(1 + x)$ is the "simple" MOND interpolating function identified empirically by Milgrom [21]. The present framework derives it from field closure rather than fitting it to data, but the resulting phenomenology is identical at galaxy scales. It is therefore important to address the observational challenges that MOND-type modifications face.

Galaxy clusters present the most significant challenge: even with the MOND correction, clusters appear to require additional mass beyond what baryons provide. The Bullet Cluster, where gravitational lensing centers are offset from the visible gas, is often cited as direct evidence for particle dark matter. Additionally, the CMB power spectrum and large-scale structure formation are well-described by ΛCDM with particle dark matter.

The $(3+1)^2$ framework differs from phenomenological MOND in three respects that bear on these challenges. First, the acceleration scale a_0 is not a free parameter here; it is derived from the topology ($k^2 G m_e / r_e^2$) and linked to the Hubble constant through $a_0 = cH/(2\pi) \cdot \sqrt{k/\pi}$. This means a_0 evolves with cosmic time as H evolves, potentially altering the dynamics at cluster scales and high redshift. Second, the framework predicts a geometric aperture correction $\sqrt{k/\pi} = 1.128$ that has no analogue in phenomenological MOND. The physical origin is a mismatch between two geometries.

When we calculate $a_0 = cH/(2\pi)$, we use π, the signature of continuous spherical geometry, circles and smooth surfaces. But gravity does not couple through a smooth sphere. It couples through $k = 4$ discrete tetrahedral faces, flat polygonal ports in a round boundary. The situation is analogous to looking at a landscape through four square windows in a circular room: the windows show a systematically different view than an unobstructed panorama, because flat openings sample a curved surface differently than the surface samples itself.

The two geometric signatures are $k = 4$ (discrete, polyhedral) and π (continuous, spherical). Their ratio $k/\pi = 4/\pi \approx 1.27$ measures the mismatch at the area level: how the flat coupling faces relate to the curved field boundary. But the Hubble constant $H = v/d$ depends on distance, not area. Since area scales as length squared, we convert to a length-scale correction by taking the square root:

$$\text{Aperture correction} = \sqrt{\frac{k}{\pi}} = \sqrt{\frac{4}{\pi}} = 1.128 \tag{54}$$

This correction applies whenever a measurement assumes continuous spherical geometry but the underlying coupling is discrete and tetrahedral. The MOND relation $a_0 = cH/(2\pi)$ makes exactly this assumption: it uses π as if gravity couples through a smooth sphere. Correcting for the actual tetrahedral coupling gives $H = 69.92$ km/s/Mpc, matching the gravitational-wave standard siren measurement (70.0 ± 12 km/s/Mpc) [12]. Third, the derivation from detailed balance (the two-rate equilibrium above) provides a physical mechanism absent in the empirical MOND prescription.

Whether these differences resolve the cluster-scale and CMB challenges remains an open question. A full treatment of cluster dynamics and structure formation within the $(3+1)^2$ framework is beyond the scope of this paper. We note the challenges here because intellectual honesty requires acknowledging them, and because addressing them is a clear target for

future work.

The resolution of galaxy-scale dark matter is conceptually identical to correcting tidal calculations by accounting for water's molecular structure. Standard cosmology ignores spacetime structure; the $(3+1)^2$ framework accounts for it. The dark matter problem at galactic scales is resolved not by finding invisible particles, but by recognizing that gravity couples through cellular topology with a characteristic scale $a_0 = k^2 G m_e / r_e^2$. Whether this resolution extends to all scales is the next question.

E. Precision Context

The preceding sections derived nine fundamental equations from the $(3+1)^2$ topology. This subsection examines the collective precision of these results. A single accurate prediction might be coincidental; systematic agreement across multiple domains suggests the topology captures real structure.

Quantity	**Equation from $(3+1)^2$ Topology**	**Observed**	**Dev.**
G	$G = \frac{(3+1)^2+1}{3(3+1)+1} \times \alpha^{1+(3+1)+(3+1)^2} \times \frac{\hbar c}{m_e^2}$	6.674×10^{-11}	0.12%
$1/\alpha$	$\frac{1}{\alpha} = 1 + 2k(k^2+1) + \frac{1}{k(k+d)} = 137 + \frac{1}{28}$	137.036	0.00021%
H	$H = \frac{2\pi a_0}{c\sqrt{k/\pi}} = \frac{78.90}{1.128}$ km/s/Mpc	70.0 ± 12	0.11%
m_e	$m_e = \left(\frac{H \cdot e^4 \cdot \sqrt{k/\pi}}{2\pi k^2 G (4\pi\varepsilon_0)^2 c^3}\right)^{1/3}$	9.109×10^{-31} kg	0.10%
a_0	$a_0 = \frac{k^2 G m_e}{r_e^2} = \frac{16 G m_e}{r_e^2}$	1.2×10^{-10} m/s^2	0.41%

TABLE VII. Equations derived from $(3+1)^2$ topology with zero adjustable parameters. Every numerical factor emerges from $(3+1)$ recursion: prefactor $[(3+1)^2+1]/[3(3+1)+1]$, exponent $21 = 1+(3+1)+(3+1)^2$, coupling $k^2 = (3+1)^2 = 16$, breadth $1+2k(k^2+1)+1/[k(k+d)] = 137.036$. Mean deviation: $0.15\% \pm 0.07\%$ (standard error).

The gravitational constant. Why does gravity couple through a 21-level hierarchy? The

$(3+1)^2$ topology provides the answer: gravitational interactions traverse three coupling levels, yielding exponent $1+(3+1)+(3+1)^2 = 21$. The prefactor $[(3+1)^2+1]/[3(3+1)+1]$ counts coupling channels over structural elements in the dual tetrahedral geometry. The predicted value deviates from measurement by 0.12%, small for a zero-parameter prediction but larger than the ∼0.05% inter-experiment scatter [22]. This residual may indicate higher-order corrections not captured by the leading-order topology.

The round-trip test. A stringent validation involves bidirectional consistency. Starting from G, α, $\hbar$, c, and m_e, the topology derives $H = 69.92$ km/s/Mpc, matching the gravitational-wave standard siren measurement [12] to 0.11%. Reversing this calculation, the observed H recovers m_e to 0.10% of the CODATA value [22]. This closure would fail if the relationships were coincidental; bidirectional consistency rules out curve-fitting.

Connection to modified gravity. The acceleration scale a_0 derived above is the same quantity that resolves the dark matter problem (§VII D). The 0.41% match between the derived a_0 and the empirical MOND scale is not coincidental: both emerge from the same $(3+1)^2$ closure structure. The precision table thus validates not only the framework's derivation of fundamental constants, but also its resolution of galactic dynamics.

Context. The relevant comparison is between frameworks. ΛCDM postulates 85% invisible mass to explain galactic dynamics; this framework derives the observed dynamics from spacetime structure while predicting G, H, m_e, and a_0 to sub-percent precision with zero adjustable parameters. The 0.12% residual in G exceeds the ∼0.05% inter-experiment scatter, suggesting either higher-order corrections or systematic effects in measurements. Whether this residual constitutes a deficiency is a question the data will resolve. The standard paradigm, by contrast, has no prediction for G at all.

F. The Frontier: The Electron Mass

Why does the electron have the mass it has?

This question has challenged physics for over a century. Dirac called mass "one of the fundamental unsolved problems." The Standard Model treats particle masses as free parameters, Yukawa couplings that must be measured rather than derived. Twenty-six such parameters define the model, each determined empirically, none explained geometrically. The electron

mass $m_e = 9.109 \times 10^{-31}$ kg appears in every branch of physics, yet no theory has derived its value from first principles.

The $(3+1)^2$ framework changes this situation.

The three-equation system

Consider what the framework has established. Closure geometry requires $k = 4$ faces (the simplex theorem). Isotropic coupling engages all $k^2 = 16$ channels. The fine structure constant α emerges from the ratio of closure rate to Compton frequency (Result 6). The Hubble constant H is observable. What remains?

Three equations constrain the size r and mass m of a single tetrahedral field region, or "tile," in the $(3+1)^2$ structure:

Equation 1: Closure saturation. The MOND acceleration scale marks where gravitational closure saturates:

$$a_0 = k^2 \frac{Gm}{r^2} \tag{55}$$

The factor $k^2 = (3+1)^2 = 16$ counts the coupling channels of the dual tetrahedral structure.

Equation 2: Cosmological connection. The observable universe has a cosmological horizon, a maximum distance from which light can reach us. Thermodynamic arguments applied to this horizon, analogous to those that relate a black hole's temperature to its surface gravity, yield a minimum acceleration scale tied to the Hubble expansion rate H. This relationship was derived in the companion paper [2]:

$$a_0 = \frac{cH}{2\pi}\sqrt{\frac{k}{\pi}} \tag{56}$$

The geometric correction $\sqrt{k/\pi} = \sqrt{4/\pi} \approx 1.128$ accounts for the difference between tetrahedral packing (the discrete structure of the field) and spherical symmetry (the continuous geometry of the horizon).

Equation 3: Quantum-electromagnetic relation. The tile's physical scale and the quantum scale are related by α:

$$m = \frac{\alpha\hbar}{rc} \tag{57}$$

This is the definition of the classical electron radius, inverted to express mass in terms of radius.

The derivation

Three equations, two unknowns (r and m). The system is overdetermined, meaning it has a unique solution if consistent.

Equating the two expressions for a_0 (Eqs. 55 and 56):

$$k^2 \frac{Gm}{r^2} = \frac{cH}{2\pi}\sqrt{\frac{k}{\pi}} \tag{58}$$

Substituting $m = \alpha\hbar/(rc)$ from Eq. 57:

$$k^2 G \cdot \frac{\alpha\hbar}{rc} \cdot \frac{1}{r^2} = \frac{cH}{2\pi}\sqrt{\frac{k}{\pi}} \tag{59}$$

Solving for r^3:

$$r^3 = \frac{2\pi k^2 \alpha\hbar G}{c^2 H\sqrt{k/\pi}} = 2\pi^{3/2}k^{3/2} \cdot \frac{\alpha\hbar G}{c^2 H} \tag{60}$$

For $k = 4$:

$$r^3 = 16\pi^{3/2} \cdot \frac{\alpha\hbar G}{c^2 H} \tag{61}$$

Result 7: The Electron Mass from Closure Geometry

$$\boldsymbol{r = \left((3+1)^2\pi^{3/2} \cdot \frac{\alpha\hbar G}{c^2 H}\right)^{1/3} = 2.82 \times 10^{-15}\ \text{m}} \tag{62}$$

$$\boldsymbol{m = \frac{\alpha\hbar}{rc} = 9.11 \times 10^{-31}\ \text{kg}} \tag{63}$$

Predicted: $\boldsymbol{r = 2.82}$ fm, $\boldsymbol{m = 9.11 \times 10^{-31}}$ kg

Measured: $r_e = 2.818$ fm, $m_e = 9.109 \times 10^{-31}$ kg

The "electron" is not an input. It is an output of closure geometry.

Why this works

Three equations constrain two unknowns. Mathematicians call such a system *overdetermined*: it contains more constraints than strictly needed to fix a solution. Most overdetermined systems have no solution at all. Asking for a number that is simultaneously the square root of 9, half of 7, and one-third of 10 yields no answer, because those constraints are incompatible.

GPS navigation relies on this principle. Three satellites suffice to fix a position in three-dimensional space, but receivers use four or more. If the extra satellite disagrees, the position is unreliable. If all satellites agree, confidence increases, precisely because the extra constraint was not guaranteed to be consistent.

The three equations above arise from different domains: gravitational closure (Eq. 55), cosmological horizon physics (Eq. 56), and quantum electrodynamics (Eq. 57). They share no common derivation. Yet they converge on a single pair of values (r, m), matching measurement to better than 0.1%. This consistency across independent domains is the mathematical signature of an underlying structure.

A note on circularity. One might ask whether this system is circular, since Eq. 57 is the classical electron radius definition inverted. The key is the input chain: the system's inputs are $k = 4$ (from the simplex theorem), G (from Result 1), α (from Result 6), $\hbar$, c, and H (a cosmological observable measured independently of particle physics, e.g., from Cepheid variables, gravitational-wave standard sirens, and CMB observations). None of these inputs assume the existence or mass of the electron. The three equations then *output* r and m, which happen to match r_e and m_e. The electron is genuinely a prediction, not an input recovered by algebraic rearrangement.

What this means

The electron mass is not a free parameter. It is determined by the requirement that gravitational closure (k^2Gm/r^2) match the cosmological acceleration scale ($cH/2\pi$) at the quantum-electromagnetic boundary ($m = \alpha\hbar/rc$). The geometry forces the mass.

The electron mass links quantum mechanics ($\hbar$), gravity (G), electromagnetism (α), and cosmology (H) through a single topological integer: $k = 4$. These domains were developed independently over four centuries. That they converge here, on a shared geometric structure, suggests the divisions between them are artifacts of history rather than nature.

This completes the framework. The Hubble constant H is measured from cosmological observations (Cepheid variables, Type Ia supernovae, gravitational-wave standard sirens) that involve no knowledge of the electron mass. With α derived from topology (Result 6) and H measured independently of particle physics, no empirical particle-physics input remains. The fundamental scales of nature, from $\hbar$ to G to m_e, emerge from $(3+1)^2$ closure geometry.

Open questions

Cosmological implications. The Field Origin, where $t = 0$, lies at the center of every tile. What does a universe built from such structures imply for the nature of time at large scales?

Experimental tests. The predicted value of G can be tested. Precision measurements continue to improve. Agreement or disagreement with the 0.12% prediction will determine whether the geometric interpretation is correct.

Higher-order corrections. The 0.12% residual in G exceeds inter-experiment scatter. Does this reflect systematic measurement effects, or does the framework require refinement?

Covariant completion. The results presented here derive from the static topology of the stellated octahedron. Two action principles bear on the dynamics.

The established Hassan-Rosen bimetric action [10], written in vielbein form for two tetrads

e^I_α (TetA) and $\tilde{e}^I_\alpha$ (TetB), is:

$$S_{\rm HR} = \int d^4x \left[e\, e^\alpha_I e^\beta_J \Omega^{IJ}_{\alpha\beta}(e) \;+\; \tilde{e}\, \tilde{e}^\alpha_I \tilde{e}^\beta_J \tilde{\Omega}^{IJ}_{\alpha\beta}(\tilde{e}) \;+\; m^2 e \sum_{n=0}^{4} \beta_n\, e_n(e^{-1}\tilde{e}) \right] \tag{64}$$

The first two terms are the independent curvatures of each tetrahedron. The third is their interaction through the elementary symmetric polynomials e_n of the frame transition matrix $e^{-1}\tilde{e}$, weighted by five parameters β_n. The total number of coupling terms across all five polynomials is $1 + 4 + 6 + 4 + 1 = 16 = (3+1)^2$, matching the bimetric coupling space of this framework. The $(3+1)^2$ topology constrains which values of β_n are realized.

A complementary scalar-tensor completion, developed in "A Covariant Theory Without Dark Matter" [3], integrates the constitutive law $\mu(x) = x/(1+x)$ directly:

$$S_{\rm ST} = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \frac{a_0^2}{8\pi G}\, \mathcal{F}\!\left(\frac{|\nabla\Phi|^2}{a_0^2} \right) + \mathcal{L}_{\rm matter} \right] \tag{65}$$

where $\mathcal{F}(y) = \frac{1}{2}y - \sqrt{y} + \ln(1+\sqrt{y})$ is the unique Lagrangian density satisfying $d\mathcal{F}/d\sqrt{y} = \sqrt{y}\,\mu(\sqrt{y})$. This action has a single metric, a scalar field Φ, and zero free parameters: the Lagrangian is entirely determined by the constitutive law.

The bimetric action (Eq. 64) provides the natural framework for two coupled tetrahedra; the scalar-tensor action (Eq. 65) provides the derived dynamics from field closure. How these two relate, whether the scalar-tensor theory emerges as the effective low-energy limit of the bimetric theory with topologically constrained β_n, is a target for future work.

G. Closing Thought

Newton measured G over three centuries ago. Sommerfeld introduced α in 1916. Planck postulated $E = hf$ in 1900. De Broglie proposed $\lambda = h/p$ in 1924. For all that time, these were separate discoveries, unconnected except by their presence in the same universe.

This paper and its predecessor propose a different view: they are projections of one geometric structure.

The stellated octahedron, the simplest stable configuration of two tetrahedra in three dimensions, connects what appeared separate. Force laws, mass-energy, the hierarchy, quantum foundations, the constants, the dark matter problem, and the electron mass itself all trace

to the same $(3+1)^2$ topology. Seven equations, a dark matter resolution, and a derivation of m_e, from one geometry.

Physical law may not be arbitrary. The constants may not be accidents. Behind the equations lies structure, and at the center of that structure lies a point where time is zero.

The investigation continues.

DATA AVAILABILITY AND SUPPLEMENTAL MATERIALS

This paper is a theoretical derivation. No experimental data were generated. All numerical values are computed from CODATA 2022 fundamental constants [22]. The following supplemental materials are available:

Complementary Technical Notes. Three self-contained technical notes develop aspects of the framework that require extended treatment:

1. *Covariant-Complete $(3+1)^2$ Gravity: A Uniquely Determined Lagrangian from Topological Field Closure.* Presents the zero-parameter scalar-tensor action derived from the $(3+1)^2$ topology, its field equations (modified Poisson, modified Friedmann, and growth equations), and five zero-parameter predictions spanning gravity, cosmology, particle physics, galactic dynamics, and the CMB.

2. *The Origin of Every Integer in G: Counting Coupling Channels in Ghost-Free Bimetric Gravity.* Traces each integer in $G = \frac{17}{13}\alpha^{21}\frac{\hbar c}{m_e^2}$ to a specific counting question about the stellated octahedron, with a term-by-term translation table mapping every integer to its GR or bimetric gravity origin. Includes the sensitivity analysis demonstrating that ± 1 in the exponent shifts G by exactly $1/\alpha = 137$.

3. *Why the Fine-Structure Constant Governs Gravitational Coupling: Two Routes to One Scale and the Derived Constitutive Law.* Establishes why α appears in the gravitational formula (electromagnetism and gravity converge on the same topological scale r_e through independent physics) and derives the constitutive law $\mu(x) = x/(1+x)$ from the coupling geometry of the stellated octahedron and detailed balance kinetics.

Computational Validation. A Python validation suite (`run_all_validations.py`) verifies all numerical claims using full-precision CODATA 2022 constants: G (0.12%), a_0

(0.41%), m_e (0.1%), H (0.11%), exponent sensitivity, fine-structure constant, spin from face ratio, rotation curve predictions, and bidirectional round-trip consistency. Standard library Python 3.6+; no external dependencies.

Figure Generation. The script `dark_matter_corrective_lens.py` generates Figure 6. Requires `numpy` and `matplotlib`.

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